Multivariate Rank-based Distribution-free Nonparametric Testing using Measure Transportation - Supplement

Nabarun Deb*

Department of Statistics, Columbia University

Bodhisattva Sen†

Department of Statistics, Columbia University

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A Real data analysis

In this section, we will provide some real data examples where we shall compare the performance of RdCov_n^2 (henceforth referred to as RdCov) with usual distance covariance, and also Pearson's correlation. For the sake of transparency, we have chosen these real data examples from benchmark data sets which have been analyzed previously in the literature. In subsequent data analysis, we have implemented the distance correlation (DCR) based test using the

*E-mail: nd2560@columbia.edu

†Supported by NSF Grant DMS-2015376; e-mail: bodhi@stat.columbia.edu

`dcor.test` function in the `energy` package, and the Pearson's correlation test (P) using the `cor.test` function in the `stats` package in `R` ([41]).

Example A.1 (US crimes data set)**.** *Consider the US crime data set from* 110 *metropolitan areas with populations larger than* $250,000$*. The particular attributes in the data set include — (i) population (in thousands, from* 1968*), (ii) nonwhite (percentage of nonwhite population,* 1960*), (iii) density (population per square mile,* 1968*) and (iv) crime (crime rate per thousand,* 1969*). This data set is available in [15].*

The data set contains missing values. In fact, complete data is available for 100 *out of the* 110 *metropolitan areas, and we will restrict all subsequent data analysis to these* 100 *metropolitan areas. This preprocessing is exactly the same as in [48]. Also, as in [48], we are interested in answering which two of the four attributes (mentioned above) are associated among each other. The following table describes our findings:*

Dataset	Nonwhite			Density			Crime		
	(P)	(DCR)	(RdCov)	(P)	(DCR)	(RdCov)	(P)	(DCR)	(RdCov)
Population	0.49	0.01	0.38	0.00	0.00	0.00	0.00	0.00	0.00
Nonwhite				0.98	0.29	0.02	0.00	0.00	0.00
Density							0.27	0.03	0.03

Table A.1: P-values for Pearson's correlation, distance correlation and rank distance covariance for the US crime data set.

In Table A.1, apart from population (i) versus nonwhite population (ii), and nonwhite population (ii) versus population density (iii), all the other possible combinations are shown to be associated (at least at the 5% *level) by all the three tests. So, let us inspect the above two possible combinations more carefully. First let us look at the scatter plot between total population and nonwhite population (top left panel in Figure A.1). It seems that there is no linear relationship between these variables, which is why Pearson's correlation does not detect any association between these variables. However, if we look at the scatter plot between the ranks of total and nonwhite populations, there does seem to be a linear relationship (see top right*

panel in Figure A.1). This is borne out by a standard linear regression analysis with nonwhite population ranks as the response and total population ranks as the covariate, which leads to the following output:

```
> summary(fit)
Call: lm(formula = var2[, 2] ~ var1[, 2])
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)          0.37175    0.05668   6.559 2.56e-09 ***
var1[, 2]  0.26386    0.09744   2.708  0.00799 **
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

This suggests a clear association between these variables which is picked up by both distance correlation (at 5%) and rank distance covariance (at 1%).

Next, let us focus on the association between nonwhite population and population density. Firstly, it is natural to expect an association in this case, given that total and nonwhite population were associated (as argued in the previous paragraph). Once again, the corresponding scatter plot (see the bottom left panel of Figure A.1) reveals no linear relationship, whereas the corresponding plot of ranks (see the bottom right panel in Figure A.1) shows some linear relationship. This is supported by standard linear regression analysis as before:

```
> summary(fit)
Call: lm(formula = var2[, 2] ~ var1[, 2])
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)          0.38992    0.05721   6.815 7.71e-10 ***
var1[, 2]  0.22789    0.09836   2.317   0.0226 *
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Once again, there is evidence of association, but in this case, only our proposed rank distance

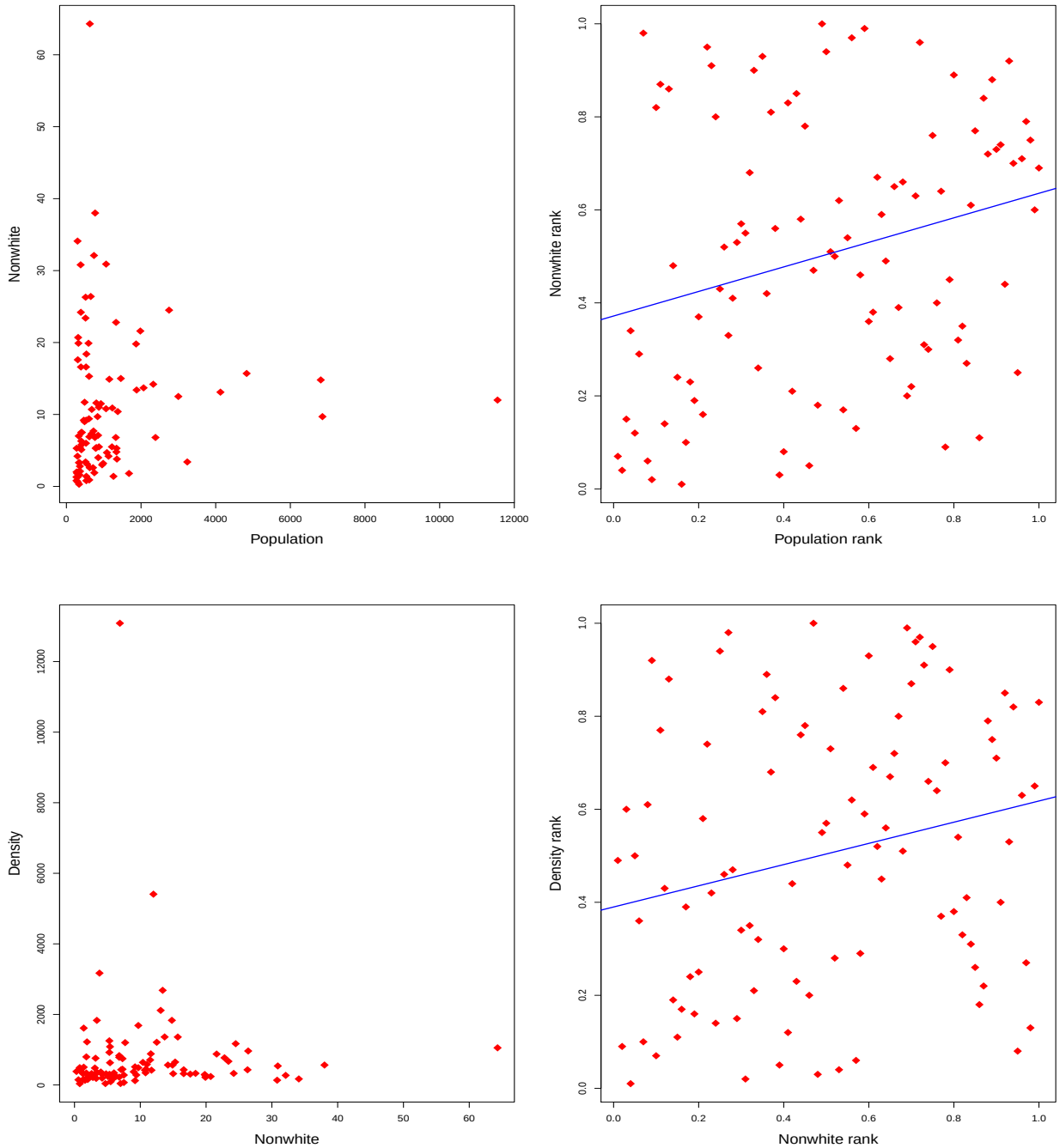

Figure A.1: The top left panel shows the scatter plot between total population (i) and nonwhite population (ii). The top right panel plots the population ranks from the 100 metropolitan areas versus the corresponding nonwhite population ranks. The bottom left panel shows the scatter plot between nonwhite population (ii) and population density (iii). The bottom right panel plots the nonwhite population ranks from the 100 metropolitan areas versus the corresponding population density ranks.

covariance test detects this association whereas Pearson's correlation and distance correlation support the null hypothesis of independence. We believe that this is because of the presence of outliers, as evidenced by the scatter plot in the bottom left panel of Figure A.1. As distance correlation requires finite moment assumptions for consistency, its performance can be affected adversely by the presence of outliers, which however, rank distance covariance is successfully robust to. This provides a real practical example when rank distance covariance may be more useful than usual distance covariance in exploratory association studies.

Remark A.1. *In the above data analysis, the performance of rank distance covariance stays the same irrespective of whether we use cutoffs from the universal distribution for fixed n or the universal limit distribution (see Theorem 4.1 in the main paper). We believe that the convergence of the fixed n universal distribution to its asymptotic limit happens rather quickly when $d_1 = d_2 = 1$. We will discuss more on this in Appendix D.4.*

Example A.2 (SONAR data set)**.** *In this example, we look at a multivariate two-sample equality of distributions testing problem based on the benchmark data set* `Sonar` *(see [37]) available in the* `R` *package* `mlbench` *The data comprises patterns obtained by reflecting sonar signals off a metal cylinder or a roughly cylindrical rock. Each pattern is a 60 dimensional vector, with each entry between 0 and 1. Each number represents the energy of the signal within specific frequency bands, integrated over time. There are 111 patterns from signals bounced off metal cylinders (M-signals) and 97 patterns from signals bounced off rocks (R-signals). As in [8], we are interested in testing whether the patterns arising out of M-signals and R-signals have the same distribution.*

Since this 60 dimensional data is difficult to visualize, we first obtained the principal components corresponding to the M-signals and then projected the patterns corresponding to both kinds of signals along those 60 directions. This results in 60 one-dimensional projections corresponding to patterns from both M-signals and R-signals. In Figure A.2, we show the QQ-plots between M-signals and R-signals corresponding to 3 of these 60 projections. It is clear in the plots that none of the distributions of the 3 one-dimensional projections are the same. In fact,

in 21 *out of these* 60 *projections we get similar QQ-plots and corresponding p-values (from a two-sample Kolmogorov-Smirnov test) are all below* 0.001*.*

The above discussion highlights that it is reasonable to expect that the null hypothesis from the two-sample equality of distributions test of interest should be rejected. At the standard 5% *level of significance, our proposed rank energy test (RE), Rosenbaum's crossmatch test (RC), the usual energy test (EN) and the Heller-Heller-Gorfine test (HHG) reject the null hypothesis, whereas the maximum mean discrepancy test (MMD) with Gaussian kernel fails to reject the null.*

B Computation of the test statistics

We begin by introducing the *assignment problem* and illustrating its connection to the computation of our proposed multivariate ranks. Suppose that n tasks are to be divided between n agents. Any agent can be assigned to perform any task, incurring some cost that may vary depending on the agent-task assignment. It is however required that all agents perform one and only one task. Under this constraint, the assignment problem seeks to find the agent-task allotment which minimizes the overall cost. Suppose that the list of agents is denoted by $\{p_1, p_2, \ldots, p_n\}$ and the list of tasks by $\{t_1, t_2, \ldots, t_n\}$. Also let $C(p_i, t_j)$ denote the cost of assigning task t_j to person p_i and finally, let $\mathcal{S}_n$ denote the set of all bijective functions from the set $\{p_1, \ldots, p_n\}$ to $\{t_1, \ldots, t_n\}$. Then the above problem may be stated as:

$$\min_{f \in \mathcal{S}_n} \sum_{i=1}^{n} C(p_i, f(p_i)). \tag{B.1}$$

This problem has been studied extensively in the combinatorial optimization literature, see e.g., [36, 6]. One of the more efficient methods to solve (B.1) above is to use the *Hungarian algorithm* (see e.g., [28]) which has worst case computational complexity $\mathcal{O}(n^3)$.

It is easy to see the connection between the assignment problem as in (B.1) and our empirical

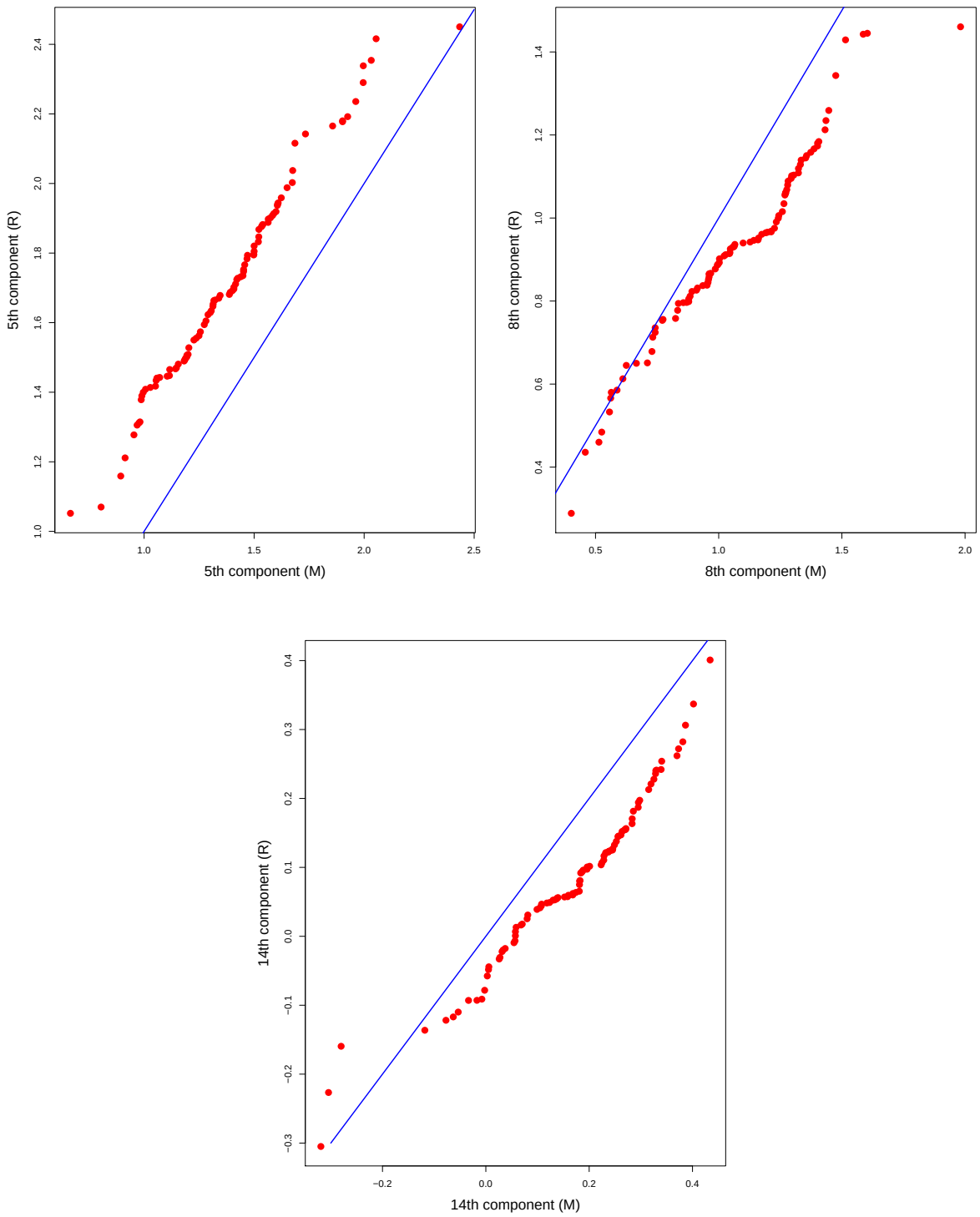

Figure A.2: QQ-plots corresponding to the one-dimensional projections of the *M-signals* and *R-signals* along the 5'th, 8'th and 14'th principal components obtained from the data matrix corresponding to the *M-signals*.

transport problem as in (2.5) in the main paper. As a result, by using the Hungarian algorithm, we can obtain the vector of multivariate ranks in at most $\mathcal{O}(n^3)$ steps. Recall that distance covariance can be computed in $\mathcal{O}(n^2 d)$ steps. Therefore, our proposed multivariate rank-based test statistic can be computed in $\mathcal{O}(n^3 + n^2 d)$ steps. A similar argument shows that our proposed multivariate rank energy statistic can be computed in $\mathcal{O}(m^3 + n^3 + mnd)$ steps.

It is important to note that the implementation of our proposed tests is extremely simple using, for example, `R`. First, we may generate the standard Halton sequence using the R-package `randtoolbox`, then we obtain the multivariate ranks using the R-package `clue` and finally we can calculate distance covariance (or the energy statistic) among the multivariate ranks using the R-package `energy`. All the methods described in this paper have been implemented using the `R` software. The relevant codes, including our simulation experiments, are available in the first author's `GitHub` page.

C Numerical experiments

In this section, we will discuss the empirical performance of our proposed tests in a wide variety of settings. Due to space constraints, we will restrict to RdCov_n^2 here; the performance of $\mathrm{RE}_{m,n}^2$ will be discussed in detail in Appendix D.1 (see the supplement).

C.1 Synthetic data experiments for mutual independence testing

We illustrate the empirical performance of RdCov_n^2 (hereafter referred to as RdCov) for testing mutual independence of two random vectors, based on synthetic data. Throughout our simulation settings, we fix $n = 200$, $d_1 = 3$ and $d_2 = 3$. In Table C.1, we compare the performance of our method with the following standard methods already existing in literature and already available in the `R` software: Pearson's correlation (P) ([40]; computed using the `stats` package), distance covariance (DCoV) ([50]; implemented in `energy` package), Hilbert-Schmidt

Independence Criteria (HSIC) ([18]; from the `dHSIC` package), mutual information (MINT) ([5]; using the `IndepTest` package), and Heller's graph-based test (HHG) ([22]; available in the `HHG` package). For MINT, we have used the `MINTav` function from the `IndepTest` package with a maximum of 20 nearest neighbors (unless otherwise specified). For the other methods, we stick to the default implementations in the quoted packages. For transparency, we have included short `R` pseudo-codes with the precise functions used in Appendix I.

As a general rule, unless otherwise specified, we construct $(\mathbf{X}, \mathbf{Y}) \in \mathbb{R}^3 \times \mathbb{R}^3$ from 3 independent random draws of $(X, Y) \in \mathbb{R} \times \mathbb{R}$. In other words $(\mathbf{X}, \mathbf{Y}) = (X_1, X_2, X_3, Y_1, Y_2, Y_3)$ where $(X_i, Y_i) \overset{i.i.d.}{\sim} (X, Y)$ for $i = 1, 2, 3$. The distributions of (X, Y) considered in this paper are as follows:

(V1) $A \sim \mathcal{N}(0,1)$, $X \sim 0.2 \times \text{Cauchy}(0,1) + A$ and $Y \sim 0.2 \times \text{Cauchy}(0,1) + A$.

(V2) $X \sim \mathcal{U}(-1,1)$ and $Y = (X^2 + \mathcal{U}(0,1))/2$.

(V3) $X \sim \mathcal{N}(0,2)$, $E \sim \text{Ber}(0.04)$, $V \sim \mathcal{N}(0,2)$ and $Y = (1-E)V + EX$. E and V are independent.

(V4) $W \sim \mathcal{U}(-1,1)$, $W_1 \sim \mathcal{U}(0,1)$, $W_2 \sim \mathcal{U}(0,1)$, $A \sim \text{Ber}(0.5)$, $V_1 = W + W_1/3$ and $V_2 = 4 \times (W^2 - 0.5)^2 + W_2$. Finally, $X = V_1$ and $Y = A \times \mathcal{N}(5,1) + (1-A)V_2$. Here W, W_1, W_2 and A are independent.

(V5) $(U_1, U_2, U_3, V_1, V_2, V_3) \sim \mathcal{N}_6(\mathbf{0}, \Sigma)$ is independent of $(W_1, W_2, W_3, Z_1, Z_2, Z_3) \sim \mathcal{N}_6(\mathbf{1}, \Sigma/2)$, where $\Sigma_{ii} = 1$ and $\Sigma_{ij} = 0.3$ if $i \leq 3, j > 3$ or $i > 3, j \leq 3$. Finally, $(X_1, X_2, X_3) \sim (1 - A_1)(U_1, U_2, U_3) + A_1(W_1, W_2, W_3)$ and $(Y_1, Y_2, Y_3) \sim (1 - A_2)(V_1, V_2, V_3) + A_2(Z_1, Z_2, Z_3)$, $A_1 \sim \text{Ber}(0.5)$ and $A_2 \sim \text{Ber}(0.3)$.

(V6) $A \sim \mathcal{N}(0,1)$, $X \sim \text{Pareto}(1,2)^2 + A$ and $Y \sim \text{Pareto}(1,1)^2 + A$.

(V7) $\epsilon \sim \mathcal{N}(0,5)$, $X \sim \mathcal{U}(0,1)$, $Y = X^{1/4} + \epsilon$. Here X and ϵ are independent.

(V8) $X \sim \mathcal{N}(0,1)$ and $Y = \log(4X^2)$.

(V9) $A \sim \mathcal{N}(0,1)$, $X \sim |A + \text{Pareto}(1,1)|^{1.5}$ and $Y \sim |A + \text{Pareto}(1,1)|^{1.5}$.

(V10) Same setup as in (V5), but with $A_2 = 0$.

Our simulation settings are similar to a variety of settings popular in the mutual independence testing literature, see e.g., [22, 9, 50]. For example, settings (V2) and (V4) are from [38] and have been used later in [22, 9]. Using mixture distributions to distinguish between different methods of independence testing is also common in literature (e.g., [22] and [9, Settings 3 and 5 from Table 1]). This is the motivation behind settings (V3), (V5) and (V10). Note that (V7) is a simple regression model; (V8) was first used in [50]. Finally, settings (V1), (V6) and (V9) have been chosen to illustrate the superior performance of RdCov when dealing with heavy-tailed distributions, as has been highlighted in the Introduction of the main paper. A representative sample of scatter plots corresponding to the simulation settings mentioned above has been provided in Figure C.1 to provide a sense of the kind of relationships portrayed in them.

In fact, the simulation settings, (V1)-(V10) used in Appendix C.1 fall broadly into three categories: (a) The dependence in $\mathbf{X}$ and $\mathbf{Y}$ comes from a mixture distribution (in (V3), (V4), (V5), and (V10)), (b) $\mathbf{Y}$ is a function of $\mathbf{X}$ with(out) an additive noise (in (V2), (V7), and (V8)), and (c) the relationship between $\mathbf{X}$ and $\mathbf{Y}$ is corrupted using a heavy-tailed distribution (in (V1), (V6), and (V9)). The scatter plots in Figure C.1 depict visuals of some of these relationships of the bivariate vector (X, Y) which represent the three categories mentioned above.

Setting (V3) in the top right panel of Figure C.1 shows a thin but distinctive mass on the $y = x$ line which makes the variables dependent. In the bottom left panel (V4) of the same figure presents a more classical mixture distribution where the relationship between X and Y

depends on the mixture component. Such settings are often used while studying graph based tests such as HHG (see [23]). In the bottom right panel of Figure C.1, for setting (V7), it seems as if there is no apparent pattern in the scatter plot. This happens because $Y = X^{1/4} + \epsilon$ where $X \sim \text{Unif}[0, 1]$ and $\epsilon \sim \mathcal{N}(0, 5)$, which implies that the noise variance is quite large compared to the signal strength. Finally the dependence between X and Y in (V9) from the top left panel is hard to detect due to contamination from a heavy-tailed Pareto noise.

With all these settings in mind, we present our findings in Table C.1. The two columns corresponding to each method represents the rejection probabilities (estimated from 1000 independent replications) at nominal levels 0.05 and 0.1.

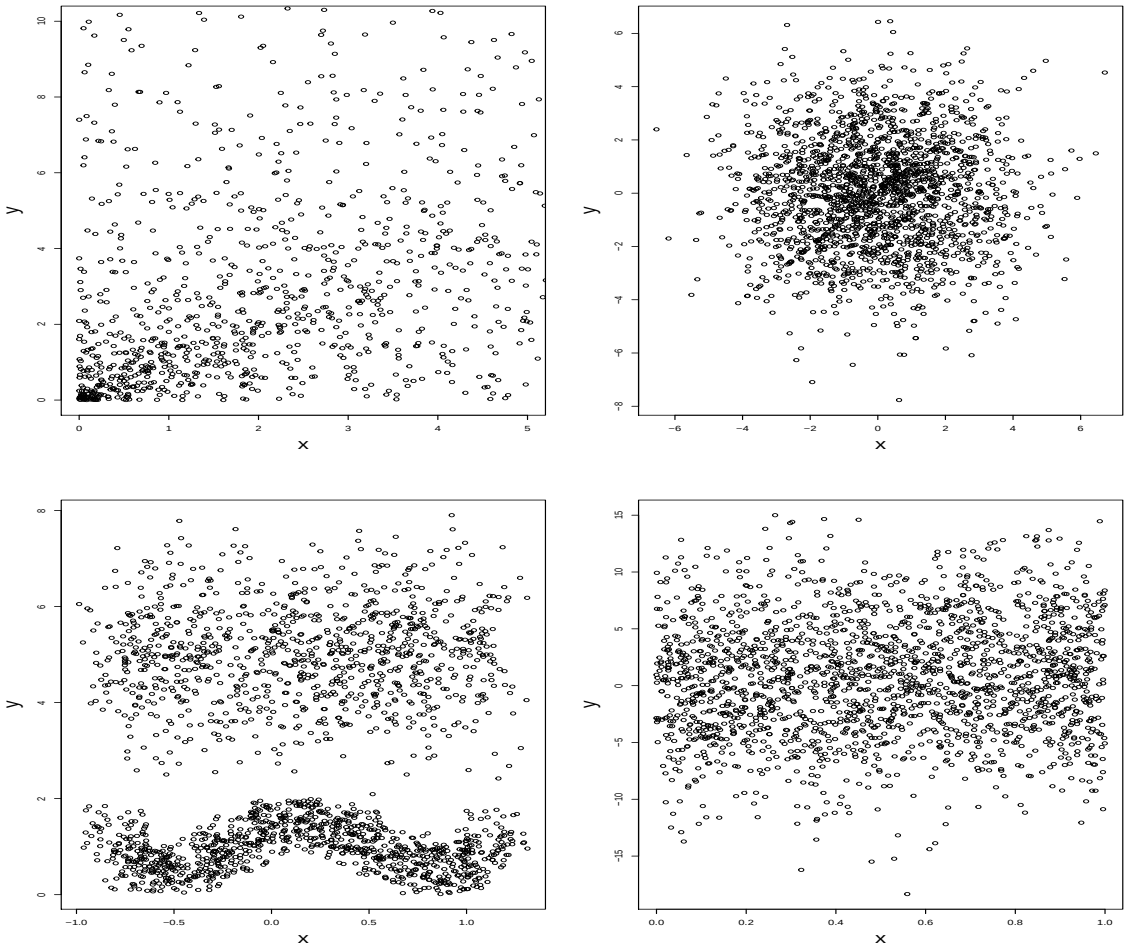

Figure C.1: Scatter plots for simulation settings from Appendix C.1. In particular, we plot setting (V9) in the top left, (V3) in the top right, (V4) in the bottom left and (V7) in the bottom right.

Below we discuss the performance of RdCov_n^2 along with the competing procedures.

(P): As one may expect, most of the methods (including ours) outperform the Pearson's correlation based testing procedure consistently. The best performances of Pearson's correlation can be observed in settings where the dependence arises largely due to a linear relationship between the variables, such as (V1), (V6) and (V10). Note that RdCov performs just as well in (V1), a lot better in (V6) and slightly worse in (V10).

	(P)		(DCoV)		(HSIC)		(MINT)		(HHG)		(RdCov)	
V1	**1.00**	**1.00**	0.81	0.60	**1.00**	**1.00**	0.93	0.84	**1.00**	**1.00**	**1.00**	**1.00**
V2	0.14	0.09	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
V3	0.15	0.08	0.15	0.08	0.15	0.09	**0.25**	**0.16**	0.14	0.09	0.14	0.07
V4	0.09	0.05	0.50	0.32	**0.95**	**0.88**	0.20	0.13	0.92	0.82	0.72	0.52
V5	0.12	0.06	0.18	0.09	0.21	0.10	0.95	0.89	**1.00**	**1.00**	0.77	0.63
V6	0.60	0.48	0.08	0.04	0.38	0.24	0.11	0.05	0.27	0.14	**0.98**	**0.95**
V7	0.25	0.16	**0.26**	**0.16**	0.22	0.13	0.11	0.06	0.19	0.10	0.21	0.13
V8	0.15	0.09	**1.00**	**1.00**	**1.00**	**1.00**	0.75	0.62	**1.00**	**1.00**	**1.00**	**1.00**
V9	0.38	0.28	0.10	0.05	0.16	0.08	0.11	0.05	0.21	0.10	**0.93**	**0.89**
V10	**0.92**	**0.86**	0.90	0.83	0.80	0.68	0.32	0.21	0.58	0.44	0.82	0.69

Table C.1: Proportion of times the null hypothesis was rejected across 10 settings. Here $n = 200$, $d_1 = d_2 = 3$. The tests with the best empirical performances have been highlighted in bold.

(DCoV): As distance covariance requires finite moments for consistency, it is expected to perform worse in the heavy-tailed settings such as (V1), (V6) and (V9). This is clearly seen in Table C.1 where RdCov outperforms usual distance covariance convincingly. Surprisingly, in (V5) where both $\mathbf{X}$ and $\mathbf{Y}$ are Gaussian mixtures, DCoV performs rather poorly whereas RdCov performs significantly better. This phenomenon can also be seen in another setting (V4) where the coordinates of $\mathbf{Y}$ arise out of a mixture distribution. Across all the other settings, namely (V2), (V3), (V7), (V8) and (V10), observe that RdCov and DCoV have almost identical performance. This leads us to believe that RdCov can be expected to perform similar to DCoV except that RdCov is a lot more robust to the presence of heavy-tailed distributions.

(HSIC): An interesting observation is that, in settings (V6) and (V9), where we have introduced a Pareto noise to Gaussian data, RdCov performs better than the test based on the HSIC (with a bounded Gaussian kernel). Note that with bounded kernels, HSIC does not need finite moment assumptions for consistency. In settings (V4) and (V5), both of which involve mixture distributions, the performance of HSIC fluctuates heavily, outperforming RdCov in (V4) and underperforming in (V5). In all the other settings, its performance is more or less

similar to RdCov.

(MINT): Table C.1 reveals that in almost all the settings RdCov outperforms MINT, except in (V3) (where all methods perform poorly) and in (V5). This could be because the consistency type results in mutual information based tests (see [5, Theorem 4]) require a number of regularity conditions which may not hold for some of the settings above. Moreover, in (V6), (V7) and (V9), MINT performs only as well as a random guess (i.e., the trivial unbiased test).

(HHG): The superior performance of our test in the heavy-tailed settings compared to competing methods persists in the case of HHG as well. This leads us to believe, in general, that RdCov is perhaps a better choice when dealing with heavy-tailed distributions than tests based on distances between data points. Note however that HHG (being based on ranks of distances) also does not require finite moment assumptions for consistency. Having said that however, the performance of HHG is definitely very competitive. Apart from the usual heavy-tailed distribution settings, the only other settings where RdCov does slightly better would be (V7) and (V10). In fact, (V10) is a slightly modified version of a Gaussian mixture model, where the dependence is mostly linear (as is supported by the superior performance of Pearson's correlation). Our general sense is that, for near-linear dependence RdCov is probably a better test than HHG. In settings (V4) and (V5) which are based on marginals having a mixture distribution, HHG convincingly outperforms RdCov (and all the other competing tests). In the rest of the settings, HHG and RdCov have similar performance.

Next we look at four very natural simulation settings based on correlated multivariate Gaussian random variables.

(IG) $(Z_1, \ldots, Z_6) \sim \mathcal{N}_6(\mathbf{0}, \Sigma)$ where $\Sigma_{i,j} = \rho \in [-1, 1]$ for $i \geq 4, j \leq 3$, $i \leq 3, j \geq 4$, $\Sigma_{i,i} = 1$ and $\Sigma_{i,j} = 0$ otherwise. Set $(X_1, X_2, X_3) = (Z_1, Z_2, Z_3)$ and $(Y_1, Y_2, Y_3) = (Z_4, Z_5, Z_6)$. Note that multivariate Gaussians are a very popular modeling choice in many practical scenarios. In [50], the authors use correlated multivariate Gaussians to compare the empirical performance of

DCoV with likelihood ratio based tests.

(IGL) $(W_1, \ldots, W_6) = (\exp(Z_1), \ldots, \exp(Z_6))$ where $(Z_1, \ldots, Z_6) \sim \mathcal{N}_6(\mathbf{0}, \Sigma)$ and $\Sigma_{i,j} = \rho \in [-1, 1]$ for $i \geq 4, j \leq 3$, $i \leq 3, j \geq 4$, $\Sigma_{i,i} = 1$ and $\Sigma_{i,j} = 0$ otherwise. Set $(X_1, X_2, X_3) = (W_1, W_2, W_3)$ and $(Y_1, Y_2, Y_3) = (W_4, W_5, W_6)$. Such log-normal distributions have often been used to understand the robustness of testing procedures, see [7].

(ISN) Let $X_1 \sim \text{Unif}[-\pi, \pi]$, $Y_1 \sim \text{Unif}[-\pi, \pi]$ where the joint density of (X_1, Y_1) is given by:

$$f(x_1, y_1) \propto 1 + \sin(\lambda x_1) \sin(\lambda y_1).$$

Here $\lambda \in \mathbb{N}$ and $\lambda \in \{1, 2, \ldots, 10\}$. Next generate $X_2, Y_2 \overset{i.i.d.}{\sim} \text{Unif}[0, 1]$. In this setting, as λ increases, the oscillations in $f(\cdot, \cdot)$ become increasingly local thereby making detection of dependence difficult. This setting was used in [5, 46].

(IST) Let $X_1 \sim \text{Unif}[-1, 1]$, $\epsilon_1 \sim \mathcal{N}(0, 1)$, ϵ_1 and X_1 are independent. Then define,

$$Y_1 = f(X_1) + 10\lambda\epsilon_1,$$

where $\lambda \in (0, 1)$ and

$$f(x) = \begin{cases} -3 & \text{if } -1 \leq x \leq -0.5 \\ 2 & \text{if } -0.5 < x \leq 0 \\ 4 & \text{if } 0 < x \leq 0.5 \\ 3 & \text{if } 0.5 < x \leq 1. \end{cases}$$

Finally draw $(X_2, Y_2), (X_3, Y_3)$ i.i.d. from the same distribution as (X_1, Y_1). This setting is motivated from the "4 clouds" setting as discussed in [38, 23].

In Figure C.2 we present plots corresponding to the power curves of different tests of independence as ρ or λ varies, as the case may be. The top left panel of Figure C.2 reveals that RdCov significantly outperforms MINT and HHG for the setting (IG). This reinforces

our belief that RdCov is a better test under near-linear dependence relations. In fact, RdCov also marginally outperforms HSIC. In setting (IG) DCoV has the best power curve, which is close to both RdCov and HSIC. Next, for setting (IGL), RdCov has the best performance as it convincingly outperforms all its competitors. In this setting, we used `MINTauto` with a maximum of 50 nearest neighbors as it yielded a marginally better performance than `MINTav` with 20 nearest neighbors. The performance of MINT is still surprising when $\rho < 0$. Although MINT should be consistent in this setting given [5, Theorem 6], we believe that the poor performance of MINT could be because the asymptotics may not have kicked in yet for $n = 200$ and $d_1 = d_2 = 3$. Note that (IGL) is a somewhat heavy-tailed setting, although the associated distribution has all moments finite (the exponential moment becomes infinite). The superior performance of RdCov in this case further strengthens our belief that RdCov can provide a better test of independence in heavy-tailed settings. For setting (ISN), MINT has the best performance which is compatible with the observation made in [5]. In this setting RdCov has the worst performance. In fact, other than MINT all the other candidates perform quite badly as the oscillations get more and more local. In setting (IST), DCoV has the best performance immediately followed by HSIC and RdCov.

Overall, we believe that these extensive simulations show that RdCov is very competitive to DCoV and HSIC in general. Further RdCov tends to have lower power compared to HHG and MINT in "wiggly" settings but higher power than these methods when the relationship between $\mathbf{X}$ and $\mathbf{Y}$ is "smooth". Also RdCov seems to be more robust than the competing procedures when dealing with heavy-tailed distributions.

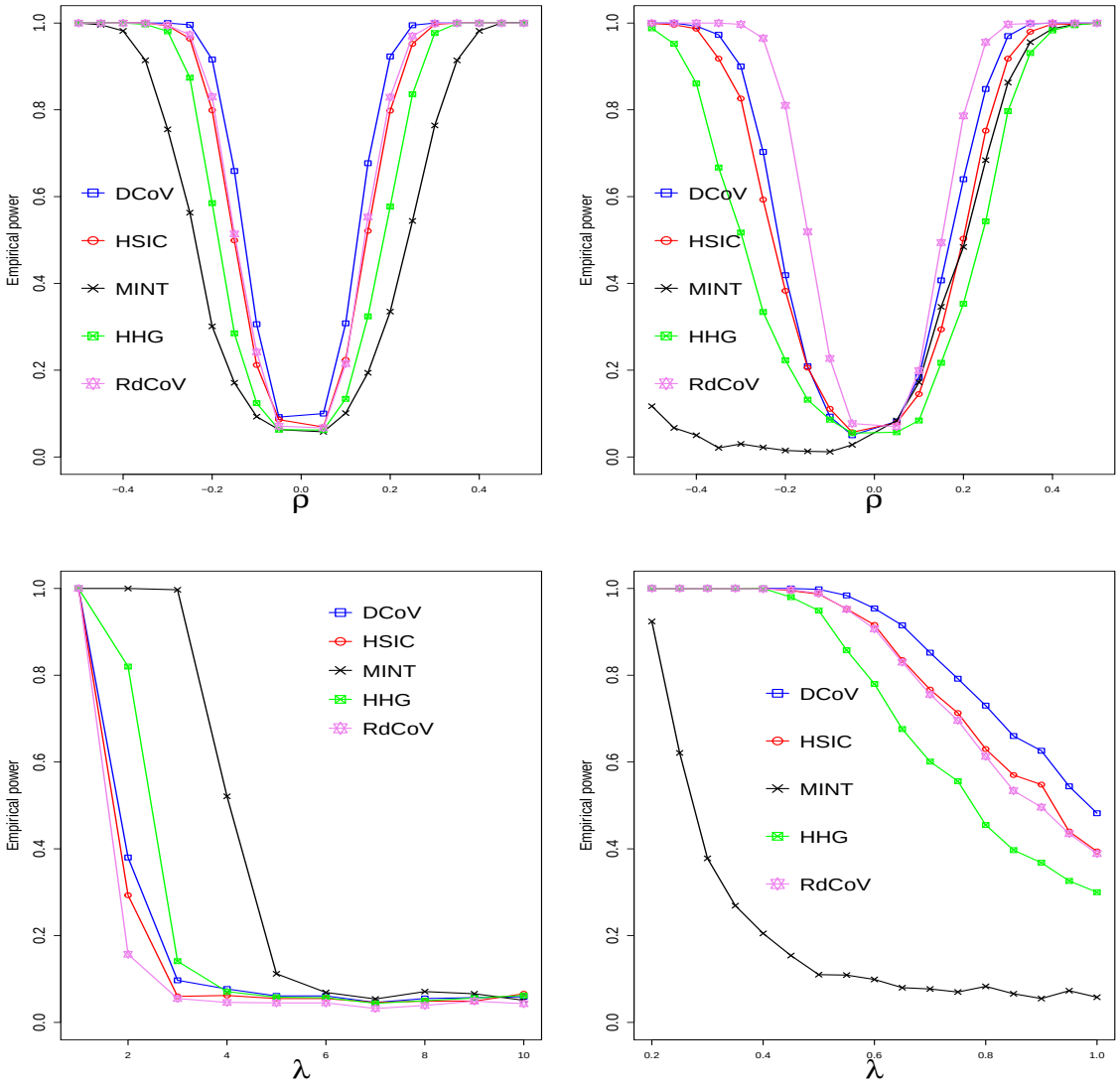

Figure C.2: The top left panel shows the power curves for (IG) at level 5% for $\rho \in [-1, 1]$. The top right panel shows the same for (IGL). The bottom right and left panels show the power curves for settings (IST) and (ISN) respectively as the parameter λ varies. The above plots are obtained by estimating the powers of aforementioned tests on a grid of appropriate parameter values using 1000 independent replicates.

D Some additional simulations

Here we continue the discussion on the empirical performance of our proposed tests from Appendix C. Among other things, we give a detailed account on the empirical performance of $\mathrm{RE}^2_{m,n}$, along with the universal asymptotic cutoffs for our proposed exactly distribution-free tests (see Tables D.8 and D.13).

D.1 Synthetic data experiments for two-sample goodness-of-fit testing

We illustrate the empirical performance of $\mathrm{RE}^2_{m,n}$ (hereafter referred to as RE) based on synthetic data. Throughout our simulation settings, we fix $m = n = 200$, and $d = 3$. We will compare our multivariate rank-based test (RE) to Rosenbaum's cross matching test (hereafter referred to as CMT; see [45]) which is the only other computationally feasible distribution-free test for two-sample goodness-of-fit testing. This test has been implemented using the `R` package `crossmatch`. In addition, we also use the usual energy test ([49]; hereafter referred to as EN) from the `energy` package, and the Heller-Heller-Gorfine test ([30]; hereafter called HHG) from the package `HHG`, as benchmarks. We also implemented the maximum mean discrepancy (MMD) based test (see [17]) using the `kmmd` function in the `R` package `kernlab` (see [29]). However its performance was convincingly poorer than the other chosen methods for our proposed simulation settings, and so we refrain from providing those details. Of course, none of HHG, EN or MMD are exactly distribution-free. All methods were implemented using the default implementations in the quoted packages.

As a general rule, we will write the three-dimensional independent vectors $\mathbf{X}$ and $\mathbf{Y}$ as (X_1, X_2, X_3) and (Y_1, Y_2, Y_3). We generate 200 i.i.d. copies of such random vectors and carry out a two-sample goodness-of-fit test based on these observations. Below we list our simulation settings:

(V1) $X_1, X_2, X_3, Y_1 \overset{i.i.d.}{\sim} \text{Cauchy}(0,1)$, and $Y_2, Y_3 \overset{i.i.d.}{\sim} \text{Cauchy}(0.2,1)$.

(V2) X_1, Y_1 are i.i.d. $\mathcal{U}^1$, $X_k = 0.25 + 0.35 \times X_{k-1} + U_k$ and $Y_k = 0.25 + 0.5 \times Y_{k-1} + V_k$ for $k = 2, 3$. Here U_2, U_3, V_2, V_3 are i.i.d. $\mathcal{U}^1$.

(V3) $\mathbf{X} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_1)$ and $\mathbf{Y} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_2)$ where $\Sigma_1(i,j) = 0.35^{|i-j|}$ and $\Sigma_2(i,j) = 0.65^{|i-j|}$ for $1 \leq i, j \leq 3$.

(V4) $\mathbf{X} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_1)$ and $\mathbf{Y} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_2)$ where $\Sigma_1(i,j) = 0.2$ for $i \neq j$ and $\Sigma_1(i,i) = 1$, $\Sigma_2(i,j) = 0.5$ for $i \neq j$ and $\Sigma_2(i,i) = 1$, for $1 \leq i, j \leq 3$.

(V5) $\mathbf{V} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_1)$ and $\mathbf{W} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_2)$ where $\Sigma_1(i,j) = 0.35^{|i-j|}$ and $\Sigma_2(i,j) = 0.75^{|i-j|}$ for $1 \leq i, j \leq 3$. Set $X_i = \exp(V_i)$ and $Y_i = \exp(W_i)$ for $i = 1, 2, 3$.

(V6) $\mathbf{V} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_1)$ and $\mathbf{W} \sim \mathcal{N}_3(\mathbf{0}, \Sigma_2)$ where $\Sigma_1(i,j) = 0.25$ for $i \neq j$ and $\Sigma_1(i,i) = 1$, $\Sigma_2(i,j) = 0.75$ for $i \neq j$ and $\Sigma_2(i,i) = 1$, for $1 \leq i, j \leq 3$. Finally, let $X_i = \exp(V_i)$ and $Y_i = \exp(W_i)$ for $i = 1, 2, 3$.

(V7) $\mathbf{X} \sim \mathcal{N}_3(\mu_1, 3\mathbf{I})$ and $\mathbf{Y} \sim \mathcal{N}_3(\mu_2, 3\mathbf{I})$ where $\mu_1 = (0,0,0)$ and $\mu_2 = (0.25, 0.25, 0.25)$.

(V8) $\mathbf{V} \sim \mathcal{N}_3(\mu_1, 3\mathbf{I})$ and $\mathbf{W} \sim \mathcal{N}_3(\mu_2, 3\mathbf{I})$ where $\mu_1 = (0,0,0)$ and $\mu_2 = (0.25, 0.25, 0.25)$. Finally, let $X_i = V_i$ and $Y_i = W_i$ for $i = 1, 2, 3$.

(V9) $X_1, X_2, X_3, V_1, V_2, V_3$ are i.i.d. Gamma$(2, 0.1)$, and W_1, W_2, W_3 are i.i.d. with the same law as $\exp(\exp(Z))$ where $Z \sim \mathcal{N}(0,1)$. Finally, set $Y_i = W_i V_i$ for $i = 1, 2, 3$.

(V10) $\mathbf{Z}_1, \mathbf{Z}_2$ are i.i.d. $\mathcal{N}_3(\mathbf{1}, \mathbf{I})$ and $A \sim \text{Ber}(0.8)$. Let $\mathbf{W} = (W_1, W_2, W_3)$ be such that W_i's are i.i.d. $\mathcal{U}(10, 11)$. Set $\mathbf{X} \coloneqq \mathbf{Z}_1$ and $\mathbf{Y} \coloneqq A\mathbf{Z}_2 + (1-A)\mathbf{W}$.

(V11) $\mathbf{Z}_1, \mathbf{Z}_2$ are i.i.d. $\mathcal{N}_3(\mathbf{1}, \mathbf{I})$ and $A \sim \text{Ber}(0.8)$. Let $\mathbf{W} = (W_1, W_2, , W_3)$ be such that W_i's are i.i.d. $\mathcal{N}(10, 0.1)$. Set $\mathbf{X} \coloneqq \mathbf{Z}_1$ and $\mathbf{Y} \coloneqq A\mathbf{Z}_2 + (1-A)\mathbf{W}$.

Many of the simulation settings above are similar to those considered in e.g., [1, 8]. For instance, setting (V2) has been adopted from [8, Section 4]. All the settings (V3)-(V8) are slightly modified versions of similar settings from [1, Section 3.3 and Appendix D]. These minor tweaks were made to make sure that the competing procedures have non-trivial power for the prescribed values of n and d. Settings (V1) and (V9) deal with scenarios where the associated distributions do not have finite first moments. In settings (V10) and (V11), we look at settings featuring mixture distributions where there is a small proportion of noise added to one of the two otherwise identical distributions. In Table D.1, we present our findings. The two columns corresponding to each methods represent the rejection probabilities (estimated using 1000 independent replications) at levels 0.05 and 0.1.

	(CMT)		(HHG)		(EN)		(RE)	
V1	0.08	0.13	0.08	0.15	0.07	0.13	**0.23**	**0.34**
V2	0.24	0.34	**0.91**	**0.94**	**0.90**	**0.94**	0.84	0.89
V3	**0.29**	0.41	0.19	0.34	0.17	0.34	**0.26**	**0.46**
V4	**0.24**	**0.34**	0.16	0.31	0.17	0.33	0.18	0.32
V5	0.61	0.73	0.54	0.70	0.35	0.56	**0.77**	**0.93**
V6	0.84	0.90	0.77	0.88	0.59	0.82	**0.96**	**0.99**
V7	0.08	0.13	0.38	0.51	**0.52**	**0.65**	0.49	0.63
V8	0.07	0.11	0.27	0.39	0.25	0.35	**0.29**	**0.43**
V9	0.06	0.06	**1.00**	**1.00**	0.97	0.97	**1.00**	**1.00**
V10	0.50	0.61	**1.00**	**1.00**	**1.00**	**1.00**	0.97	0.99
V11	0.47	0.60	**1.00**	**1.00**	**1.00**	**1.00**	0.95	0.98

Table D.1: Proportion of times the null hypothesis was rejected across 11 settings. Here $m = n = 200$ and $d = 3$.

(CMT): Table D.1 reveals a number of interesting points of comparison between CMT and RE. Let us start with settings (V10) and (V11). In these settings $\mathbf{X}$ is multivariate Gaussian, whereas $\mathbf{Y}$ is the same multivariate Gaussian with a small fractions of Uniform and Gaussian noise (20%) respectively. In both these settings RE outperforms CMT. This shows that RE is

perhaps more robust to outliers than CMT (as mentioned in the Introduction). Next, let us look at the heavy-tailed settings (V1) and (V9) (no finite first moments). Here again RE convincingly outperforms CMT, once again reinforcing the superior performance of our proposed multivariate rank-based tests in the absence of finite moments (as mentioned in the Introduction). Across settings (V3)-(V8) (based on multivariate normals and log-normals), once again RE largely outperforms CMT, except perhaps in settings (V3) and (V4). These settings feature different orders of decaying correlation and equicorrelation respectively, in a multivariate Gaussian setting. The performances of RE and CMT are almost indistinguishable in these two scenarios; in fact, CMT perhaps marginally outperforms RE in (V4). As multivariate normals and log-normals are popular modeling choices in practice, we will look into such simulation settings in greater detail in Appendix D.2. Finally, in (V2), which has the same flavor as a first order autoregressive model, once again RE significantly outperforms CMT. Overall, RE has a much superior performance than CMT in the proposed simulation settings.

(EN): The most striking feature from Table D.1 in terms of comparison between RE and EN is perhaps that RE largely outperforms EN in settings (V3), (V5), (V6) and (V8), whereas in settings (V4) and (V7) the performances are mostly comparable, EN being marginally better. This has been a recurrent observation in our simulations that the EN test (or equivalently DCoV for independence testing) loses out to RE (equivalently RdCov) when dealing with multivariate log-normals [(V4), (V6), (V8)] whereas it performs comparably when dealing with multivariate Gaussian (or mixtures thereof). This observation has been studied in more details in Appendix D.2 using multivariate normal and log-normal location alternatives. In the heavy-tailed settings, as expected, RE outperforms EN; convincingly in (V1) and marginally in (V9). In settings (V2), (V10) and (V11), EN and RE seem to have almost identical performance; EN being marginally superior. This perhaps shows that the performance of EN is also somewhat robust to small proportions of noise in data.

(HHG): Across settings (V3)-(V8) all of which are based on multivariate normal and log-normal based alternatives, RE convincingly and consistently outperforms HHG. A similar observation can be seen in the heavy-tailed setting (V1). In all the other settings, the two tests have comparable performance. In particular, being based on the ranks of pairwise distances, HHG too is perhaps somewhat robust to small proportions of outliers (as indicated by its performance in settings (V10) and (V11)).

D.2 Multivariate normal and log-normal settings

Multivariate normal and log-normal settings have been used in the context of comparing non-parametric testing procedures, e.g., [1]. In Appendix C, we used multivariate normal and log-normal settings for comparing RdCov with competing procedures for multivariate independence testing. A similar exercise can be carried out for the two-sample goodness-of-fit testing problem as well. We will consider $m = n = 200$, $d = 3$ and consider the following two settings:

(TG) $(X_1, X_2, X_3) \sim \mathcal{N}_3((\mu, \mu, \mu), 3\mathbf{I})$ where the mean parameter μ varies in $[-1, 1]$, whereas the distribution of $(Y_1, Y_2, Y_3) \sim \mathcal{N}(\mathbf{0}, 3\mathbf{I})$ stays fixed.

(TGL) $(X_1, X_2, X_3) = (\exp(Z_1), \exp(Z_2), \exp(Z_3))$ where $(Z_1, Z_2, Z_3) \sim \mathcal{N}_3((\mu, \mu, \mu), 3\mathbf{I})$ and the mean parameter μ varies in $[-1, 1]$, whereas the distribution of $(Y_1, Y_2, Y_3) = (\exp(W_1), \exp(W_2), \exp(W_3))$, with $(W_1, W_2, W_3) \sim \mathcal{N}(\mathbf{0}, 3\mathbf{I})$, stays fixed.

In Tables D.2 and D.3, we present the estimated powers (using 1000 independent replicates) for CMT, HHG, EN and RE, at levels 0.05 and 0.1. Note that, for both settings and, RE convincingly outperforms Rosenbaum's crossmatch test, which is the only other exactly distribution-free test for multivariate two-sample goodness-of-fit testing. In Table D.2, the energy test has the best empirical performance, closely followed by RE. In this setting, RE consistently outperforms HHG, rather convincingly for small values of $|\mu|$. Next, for setting RE has the best performance as it outperforms all its competitors. Note that is a somewhat

Mean	(CMT)		**(EN)**		(HHG)		(RE)	
-0.60	0.33	0.44	**1.00**	**1.00**	1.00	1.00	1.00	1.00
-0.52	0.23	0.32	**1.00**	**1.00**	0.98	0.99	0.99	1.00
-0.45	0.18	0.26	**0.98**	**0.99**	0.92	0.96	0.97	0.98
-0.37	0.14	0.20	**0.89**	**0.93**	0.78	0.86	0.85	0.91
-0.30	0.10	0.17	**0.70**	**0.79**	0.52	0.66	0.65	0.76
-0.22	0.07	0.14	**0.43**	**0.57**	0.31	0.43	0.39	0.52
-0.15	0.06	0.10	**0.20**	**0.32**	0.15	0.23	0.19	0.28
-0.07	0.05	0.09	**0.09**	**0.15**	0.08	0.14	0.08	0.15
0.07	0.06	0.10	**0.08**	**0.15**	0.07	0.14	0.08	0.15
0.15	0.06	0.11	**0.21**	**0.33**	0.15	0.25	0.19	0.30
0.22	0.08	0.12	**0.41**	**0.54**	0.28	0.41	0.38	0.50
0.30	0.09	0.14	**0.71**	**0.80**	0.50	0.66	0.64	0.76
0.37	0.14	0.22	**0.89**	**0.94**	0.73	0.84	0.84	0.92
0.45	0.17	0.26	**0.97**	**0.99**	0.90	0.95	0.96	0.98
0.52	0.25	0.36	**0.99**	**1.00**	0.98	0.99	0.99	1.00
0.60	0.33	0.45	**1.00**	**1.00**	0.99	1.00	1.00	1.00

Table D.2: Proportion of times the null hypothesis was rejected across different values of correlations for the multivariate log-normal setting (TG). In this case, EN (in bold) has the best empirical performance.

heavy-tailed setting, although the associated distribution has all moments finite (the exponential moment becomes infinite). The superior performance of RE in this case further strengthens our belief that RE can provide a better multivariate goodness-of-fit test in heavy-tailed settings.

D.3 Connection with distance covariance for bivariate normal distribution

In [50, Theorem 7], the authors explicitly calculate the population distance correlation when (X, Y) follows a bivariate Gaussian distribution with mean vector $\mathbf{0}$, variances 1 and correlation ρ. Their result interestingly shows that the population distance correlation is a strictly increasing function of $|\rho|$. In this subsection, the goal is to inspect if analogous properties hold for the rank distance correlation, defined below.

Mean	(CMT)		(EN)		(HHG)		**(RE)**	
-0.80	0.53	0.65	1.00	1.00	1.00	1.00	**1.00**	**1.00**
-0.6	0.30	0.39	0.98	0.99	0.98	0.99	**0.99**	**1.00**
-0.52	0.18	0.27	0.93	0.96	0.95	0.98	**0.99**	**1.00**
-0.45	0.16	0.25	0.85	0.91	0.89	0.94	**0.95**	**0.98**
-0.37	0.11	0.18	0.69	0.79	0.74	0.84	**0.83**	**0.90**
-0.30	0.10	0.16	0.49	0.59	0.53	0.64	**0.61**	**0.73**
-0.22	0.08	0.12	0.31	0.41	0.32	0.44	**0.35**	**0.48**
-0.15	0.06	0.11	0.15	0.24	0.17	0.26	**0.17**	**0.26**
-0.07	0.07	0.11	0.07	0.15	0.09	0.15	**0.08**	**0.16**
0.07	0.05	0.09	0.07	0.13	0.07	0.14	**0.08**	**0.14**
0.15	0.06	0.10	0.16	0.24	0.16	0.26	**0.17**	**0.28**
0.22	0.07	0.12	0.28	0.41	0.30	0.44	**0.36**	**0.49**
0.30	0.09	0.14	0.50	0.62	0.53	0.66	**0.60**	**0.72**
0.37	0.12	0.17	0.70	0.80	0.74	0.83	**0.82**	**0.88**
0.45	0.16	0.25	0.86	0.92	0.88	0.93	**0.95**	**0.98**
0.52	0.22	0.29	0.95	0.97	0.96	0.98	**0.98**	**0.99**
0.60	0.27	0.38	0.98	0.99	0.99	0.99	**1.00**	**1.00**

Table D.3: Proportion of times the null hypothesis was rejected across different values of correlations for the multivariate log-normal setting (TGL). In this case, RE (in bold) has the best empirical performance.

Definition D.1 (Rank distance correlation)**.** *The rank distance correlation* (RdCorr) *between* $\mathbf{Z}_1$ *and* $\mathbf{Z}_2$ *is defined as the usual distance correlation (see [50, Equation 2.7]) between* $R_1(\mathbf{Z}_1)$ *and* $R_2(\mathbf{Z}_2)$ *(where* $R_1(\cdot)$ *and* $R_2(\cdot)$ *are as in Definition 3.1 in the main paper). In other words,*

$$\mathrm{RdCorr}^2(\mathbf{Z}_1, \mathbf{Z}_2) := \frac{\mathrm{RdCov}^2(\mathbf{Z}_1, \mathbf{Z}_2)}{\mathrm{RdCov}(\mathbf{Z}_1, \mathbf{Z}_1)\mathrm{RdCov}(\mathbf{Z}_2, \mathbf{Z}_2)}. \tag{D.1}$$

Note that (D.1) is well-defined by Lemma 3.1 (part (c)) in the main paper. By [50, Theorem 3], it follows directly that $\mathrm{RdCorr}(\mathbf{Z}_1, \mathbf{Z}_2) \in [0, 1]$. For our proposed rank distance correlation measure (population version) above, however, a closed form expression is rather difficult to obtain when (X, Y) follows a bivariate normal distribution. But, note that the rank maps corresponding to the marginals X and Y are exactly the Gaussian cumulative distribution

functions (as discussed in Section 2.1 in the main paper). Therefore, we can use Monte-Carlo approximations with this known rank map and obtain the population rank correlation measure. The subsequent plot reveals something interesting:

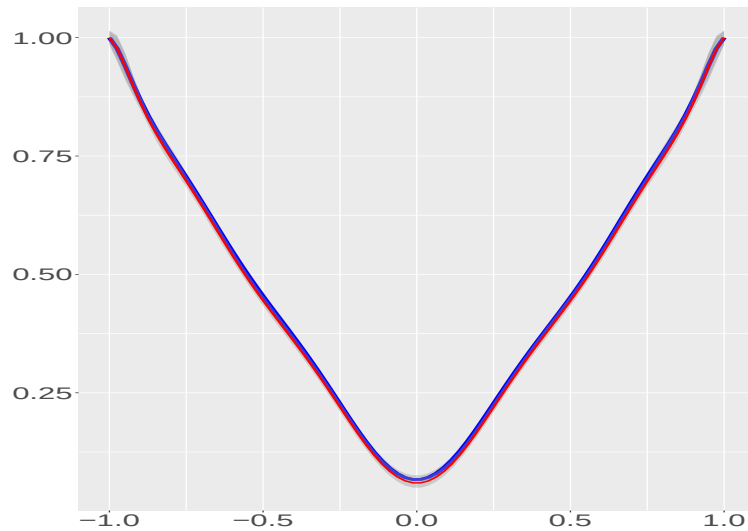

Figure D.1: The plot depicts the population distance correlation and the population rank distance correlation (y-axis) in blue and red respectively, as a function of ρ (x-axis) in the bivariate Gaussian setting.

Note that Figure D.1 shows (numerically) that the rank distance correlation is also an increasing function of $|\rho|$. The most striking feature about Figure D.1 is the proximity between the usual distance correlation and the rank distance correlation in the bivariate Gaussian case. As of now, we do not have any explanation for this striking feature. We believe that exploring the connections between rank and usual distance correlation could be an interesting area for future research.

D.4 Universal asymptotic cutoffs

Recall that the nonparametric tests proposed in Section 4 (in the main paper) have exact distribution-freeness. Therefore, it is natural to ask: "under the null hypothesis, how fast

does this fixed sequence of finite sample distributions converge to its asymptotic limit?". This question is of immense practical interest. This would help practitioners avoid using different cutoffs for different sample sizes provided the sample size is "large enough". As of now we do not have a theoretical answer to this question. In this section therefore, we attempt to answer this question through numerical experiments.

Let us start with $n\mathrm{RdCov}_n^2$. Below we present the p-values from a two-sample Kolmogorov-Smirnov test for equality of distributions, Kolmogorov-Smirnov distance (KSD) and total variation distance (TVD) between $n\mathrm{RdCov}_n^2$ and $1000\mathrm{RdCov}_{1000}^2$ as n varies, for $d_1 = d_2 = 2$ and $d_1 = d_2 = 8$ respectively.

$n =$	100	300	500	700	900
p-value	0.24	0.31	0.22	0.72	0.83
KSD	0.05	0.04	0.05	0.03	0.03
TVD	0.11	0.09	0.10	0.07	0.04

Table D.4: P-values of two-sample Kolmogorov-Smirnov tests, KSD, and TVD, when comparing $n\mathrm{RdCov}_n^2$ for $n = 100, 300, 500, 700, 900$ with $n = 1000$, for $d_1 = d_2 = 2$. All the measures are estimated using a 1000 independent replicates.

$n =$	100	300	500	700	900
p-value	0.00	0.20	0.83	0.89	0.29
KSD	0.11	0.05	0.03	0.03	0.05
TVD	0.19	0.10	0.06	0.06	0.06

Table D.5: P-values of two-sample Kolmogorov-Smirnov tests, KSD, and TVD, when comparing $n\mathrm{RdCov}_n^2$ for $n = 100, 300, 500, 700, 900$ with $n = 1000$, for $d_1 = d_2 = 8$. All the measures are estimated using a 1000 independent replicates.

Tables D.4 and D.5 show that for $d_1 = d_2 = 2$, the distribution of $n\mathrm{RdCov}_n^2$ stabilizes by $n = 300$ whereas for $d_1 = d_2 = 8$, the distribution of $n\mathrm{RdCov}_n^2$, as expected, takes longer to stabilize, potentially at around $n = 700$. Note that, in practice, it may be more useful to see when the critical values at 5% and 10% levels stabilize as that is the question practitioners may

be more interested in. Tables D.6 and D.7 provide the 5% and 10% cutoffs for the distribution of $n\mathrm{RdCov}_n^2$ as n varies, for $d_1 = d_2 = 2$ and $d_1 = d_2 = 8$ respectively. These tables show

$n =$	100	300	500	700	900
$\alpha = 0.05$	0.39	0.40	0.39	0.40	0.40
$\alpha = 0.1$	0.36	0.36	0.36	0.36	0.36

Table D.6: Thresholds for $n\mathrm{RdCov}_n^2$ when $\alpha = 0.05$, 0.1 and $n = 100, 300, 500, 700, 900$, $d_1 = d_2 = 2$. All the thresholds are estimated using a 1000 independent replicates.

$n =$	100	300	500	700	900
$\alpha = 0.05$	1.37	1.38	1.38	1.38	1.38
$\alpha = 0.1$	1.34	1.35	1.35	1.35	1.35

Table D.7: Thresholds for $n\mathrm{RdCov}_n^2$ when $\alpha = 0.05$, 0.1 and $n = 100, 300, 500, 700, 900$, $d_1 = d_2 = 8$. All the thresholds are estimated using a 1000 independent replicates.

that the 5% and 10% quantiles probably stabilize much faster, by sample size $n = 300$ even when $d_1 = d_2 = 8$. The same observation recurs for other choices of d_1, d_2 if both are less then or equal to 8. This allows us to provide universal asymptotic cutoffs as long as $d_1, d_2 \leq 8$ (see Table D.8). One can of course make such statements for other choices of d_1 and d_2.

	$d_2 = 1$	$d_2 = 2$	$d_2 = 3$	$d_2 = 4$	$d_2 = 5$	$d_2 = 6$	$d_2 = 7$	$d_2 = 8$
$d_1 = 1$	0.23	0.30	0.35	0.38	0.41	0.44	0.47	0.49
$d_1 = 2$	0.30	0.40	0.47	0.53	0.58	0.63	0.66	0.70
$d_1 = 3$	0.35	0.47	0.56	0.63	0.70	0.76	0.81	0.86
$d_1 = 4$	0.38	0.53	0.63	0.72	0.80	0.87	0.93	0.99
$d_1 = 5$	0.41	0.58	0.70	0.80	0.89	0.97	1.04	1.10
$d_1 = 6$	0.44	0.63	0.76	0.87	0.97	1.05	1.13	1.20
$d_1 = 7$	0.47	0.66	0.81	0.93	1.04	1.13	1.21	1.30
$d_1 = 8$	0.49	0.70	0.86	0.99	1.10	1.20	1.30	1.38

Table D.8: Asymptotic thresholds for $n\mathrm{RdCov}_n^2$ when $\alpha = 0.05$ and $d_1, d_2 \leq 8$. All the thresholds are estimated using a 1000 independent replicates.

We observe a similar phenomenon for $mn(m+n)^{-1}\mathrm{RE}_{m,n}^2$; see Tables D.9, D.10, D.11 and D.12. We provide the asymptotic cutoffs corresponding to

our proposed test for $1 \leq d \leq 8$ in Table D.13.

$m = n =$	100	300	500	700	900
p-value	0.22	0.61	0.83	0.65	0.69
KSD	0.05	0.03	0.03	0.03	0.03
TVD	0.09	0.07	0.08	0.07	0.06

Table D.9: P-values of two-sample Kolmogorov-Smirnov tests, KSD, and TVD, when comparing $mn(m+n)^{-1}\mathrm{RE}^2_{m,n}$ for $m = n = 100, 300, 500, 700, 900$ with $m = n = 1000$, for $d = 2$. All the measures are estimated using a 1000 independent replicates.

$m = n =$	100	300	500	700	900
p-value	0.76	0.29	0.26	0.50	0.69
KSD	0.03	0.04	0.04	0.03	0.03
TVD	0.08	0.09	0.09	0.09	0.08

Table D.10: P-values of two-sample Kolmogorov-Smirnov tests, KSD, and TVD, when comparing $mn(m+n)^{-1}\mathrm{RE}^2_{m,n}$ for $m = n = 100, 300, 500, 700, 900$ with $m = n = 1000$, for $d = 8$. All the measures are estimated using a 1000 independent replicates.

$m = n =$	100	300	500	700	900
$\alpha = 0.05$	1.09	1.17	1.15	1.16	1.16
$\alpha = 0.1$	0.93	0.95	0.89	0.90	0.93

Table D.11: Thresholds for $mn(m+n)^{-1}\mathrm{RE}^2_{m,n}$ when $\alpha = 0.05$, 0.1 and $m = n = 100, 300, 500, 700, 900$, $d = 2$. All the thresholds are estimated using a 1000 independent replicates.

E Some additional discussion

We now provide some natural extensions of the proposed testing procedures to multivariate and multi-sample settings. We will also compare our proposals for defining ranks and associated test statistics with some competing proposals.

$m = n =$	100	300	500	700	900
$\alpha = 0.05$	1.69	1.65	1.73	1.66	1.73
$\alpha = 0.1$	1.53	1.49	1.53	1.55	1.51

Table D.12: Thresholds for $mn(m+n)^{-1}\mathrm{RE}^2_{m,n}$ when $\alpha = 0.05$, 0.1 and $m = n = 100, 300, 500, 700, 900$, $d = 8$. All the thresholds are estimated using a 1000 independent replicates.

d =	1	2	3	4	5	6	7	8
	0.94	1.15	1.26	1.37	1.45	1.54	1.61	1.73

Table D.13: Asymptotic thresholds for $mn(m+n)^{-1}\mathrm{RE}^2_{m,n}$ when $\alpha = 0.05$, $d \leq 8$ and $m = n = 1000$. All the thresholds are estimated using a 1000 independent replicates.

E.1 Extension to multivariate multi-sample (≥ 2) independence and equality of distributions testing

In this section we discuss how one can construct distribution-free procedures for testing: **(I)** the mutual independence of two or more random vectors, and **(II)** the equality of two or more multivariate distributions.

(i) Testing for mutual independence of K random vectors: Suppose that we have observed i.i.d. data $\mathbf{X}_1, \ldots, \mathbf{X}_n$ from some distribution μ on $\mathbb{R}^d$. For each $1 \leq i \leq n$, suppose that $\mathbf{X}_i = (\mathbf{X}_i^1, \ldots, \mathbf{X}_i^K)$ where $\mathbf{X}_1^j \sim \mu_j \in \mathcal{P}_{ac}(\mathbb{R}^{d_j})$, where $K \geq 2$. We are interested in testing the following hypothesis (of mutual independence):

$$\mathrm{H}_0 : \mu = \mu_1 \otimes \mu_2 \otimes \ldots \otimes \mu_K \qquad \text{versus} \qquad \mathrm{H}_1 : \text{ Not } \mathrm{H}_0. \tag{E.1}$$

Before proposing the test, let us start with some notation. Fix $1 \leq j \leq K$, $1 \leq i \leq n$, and define $\mathbf{X}^j := \{\mathbf{X}_1^j, \ldots, \mathbf{X}_n^j\}$ — the n data points for the j'th marginal, $\mathbf{X}_i^{j+} := (\mathbf{X}_i^{j+1}, \ldots, \mathbf{X}_i^K)$ — the i'th data point from the $(j+1)$'th marginal variable, and $\mathbf{X}^{j+} := \{\mathbf{X}_1^{j+}, \ldots, \mathbf{X}_n^{j+}\}$ — the collection of all sub-vectors from the j'th marginal. As before, define $\mathcal{H}_n^{d_j}$ to be the (fixed) sample multivariate ranks, $j = 1, 2, \ldots, K$. Also let $\widehat{\mathbf{R}}_n^j(\cdot)$ denote the empirical rank map

which transports the empirical distribution on $\mathbf{X}^j$ to that on $\mathcal{H}_n^{d_j}$ (see (2.5) in the main paper), for $j = 1, \ldots, K$. Finally, set $\widehat{\mathbf{R}}_n(\mathbf{X}_i^{j+}) := \big(\widehat{\mathbf{R}}_n^{j+1}(\mathbf{X}_i^{j+1}), \ldots, \widehat{\mathbf{R}}_n^{K}(\mathbf{X}_i^{K})\big)$ which is a vector of dimension $d_{j+1} + d_{j+2} + \ldots + d_K$.

Now, we are in a position to define our test statistic RdCov_n^2 which is given by:

$$\mathrm{RdCov}_n^2 := \sum_{j=1}^{K-1} \mathrm{RdCov}_{n,j}^2$$

where

$$\begin{aligned}\mathrm{RdCov}_{n,j}^2 := \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\widehat{\mathbf{R}}_n^j(\mathbf{X}_k^j) - \widehat{\mathbf{R}}_n^j(\mathbf{X}_l^j)\|\right) \times \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\widehat{\mathbf{R}}_n^{j+}(\mathbf{X}_k^{j+}) - \widehat{\mathbf{R}}_n^{j+}(\mathbf{X}_l^{j+})\|\right) \\ + \frac{1}{n^2}\sum_{k,l=1}^{n} \|\widehat{\mathbf{R}}_n^j(\mathbf{X}_k^j) - \widehat{\mathbf{R}}_n^j(\mathbf{X}_l^j)\| \|\widehat{\mathbf{R}}_n^{j+}(\mathbf{X}_k^{j+}) - \widehat{\mathbf{R}}_n^{j+}(\mathbf{X}_l^{j+})\| \\ - \frac{2}{n^3}\sum_{k,l,i=1}^{n} \|\widehat{\mathbf{R}}_n^j(\mathbf{X}_k^j) - \widehat{\mathbf{R}}_n^j(\mathbf{X}_l^j)\| \|\widehat{R}_n^{j+}(\mathbf{X}_k^{j+}) - \widehat{R}_n^{j+}(\mathbf{X}_i^{j+})\|. \qquad \text{(E.2)}\end{aligned}$$

Note that the right hand side of (E.2) can be interpreted as the usual distance covariance of the transformed data points $\big\{(\widehat{\mathbf{R}}_n^j(\mathbf{X}_i^j), \widehat{\mathbf{R}}_n^{j+}(\mathbf{X}_i^{j+})) : 1 \le i \le n\big\}$, for $j = 1, \ldots, K-1$.

We *reject* the null hypothesis in (E.1) if $n\mathrm{RdCov}_n^2$ is larger than c_n where

$$c_n := \inf\{c : \mathbb{P}_{\mathrm{H}_0}(n\mathrm{RdCov}_n^2 \ge c) \le \alpha\};$$

here α is the usual prespecified type-I error level. Proposition E.1 below shows that c_n can be obtained in a distribution-free fashion just as in the $K = 2$ case (see Section 4.1 in the main paper); it also proves the consistency of the proposed test (see Appendix F.14 for the proof).

Proposition E.1. *If the marginals μ_j's are Lebesgue absolutely continuous, then, under H_0, RdCov_n^2 is a distribution-free statistic. Additionally, if the empirical distributions on $\mathcal{H}_n^{d_j}$ converge weakly to $\mathcal{U}^{d_j}$, for $j = 1, 2, \ldots, K$, then the above test which rejects H_0 when $n\mathrm{RdCov}_n^2 \ge c_n$ is consistent.*

This extension from the case $K = 2$ (in Section 4.1 in the main paper) to $K \geq 2$ (in this section) is based on the same framework as laid out in [33].

(ii) Testing for K-sample equality of distributions: Suppose that we have observed independent data $\{\mathbf{X}_i^j : 1 \leq j \leq K, 1 \leq i \leq n_j\}$, where $\mathbf{X}_1^j, \ldots, \mathbf{X}_{n_j}^j \sim \mu_j \in \mathcal{P}_{ac}(\mathbb{R}^d)$, $\sum_{j=1}^K n_j = n$, and $K \geq 2$. In this setting, we are interested in testing the following hypothesis:

$$\mathrm{H}_0 : \mu_1 = \mu_2 = \ldots = \mu_K \qquad \text{versus} \qquad \mathrm{H}_1 : \text{ Not } \mathrm{H}_0. \tag{E.3}$$

Let $\widehat{\mathbf{R}}_n(\cdot)$ denote the empirical rank map that transports the empirical distribution on the pooled sample (all $\mathbf{X}_i^j$'s taken together) to that on $\mathcal{H}_n^d$ (fixed sample multivariate ranks). We then construct our test statistic as:

$$\mathrm{RE}^2_{1:K,n} := \sum_{j=1}^{K-1} \mathrm{RE}^2_{j:(j+1),n}$$

where

$$\mathrm{RE}^2_{j:(j+1),n} := \frac{2}{n_j n_{j+1}} \sum_{m=1}^{n_j} \sum_{l=1}^{n_{j+1}} \|\widehat{\mathbf{R}}_n(\mathbf{X}_m^j) - \widehat{\mathbf{R}}_n(\mathbf{X}_l^{j+1})\| - \frac{1}{n_j^2} \sum_{m,l=1}^{n_j} \|\widehat{\mathbf{R}}_n(\mathbf{X}_m^j) - \widehat{\mathbf{R}}_n(\mathbf{X}_l^j)\| - \frac{1}{n_{j+1}^2} \sum_{m,l=1}^{n_{j+1}} \|\widehat{\mathbf{R}}_n(\mathbf{X}_m^{j+1}) - \widehat{\mathbf{R}}_n(\mathbf{X}_l^{j+1})\|. \tag{E.4}$$

Note that the right hand side of (E.4) can be interpreted as the usual energy statistic between the transformed data points $\left\{(\widehat{\mathbf{R}}_n(\mathbf{X}_i^j) : 1 \leq i \leq n_j\right\}$ and $\left\{\widehat{\mathbf{R}}_n(\mathbf{X}_i^{j+1})) : 1 \leq i \leq n_{j+1}\right\}$, for $j = 1, \ldots, K-1$.

We *reject* the null hypothesis in (E.3) if $n\mathrm{RE}^2_{1:K,n}$ is larger than c_n where $c_n := \inf\{c : \mathbb{P}_{\mathrm{H}_0}(n\mathrm{RE}^2_{1:K,n} \geq c) \leq \alpha\}$. Proposition E.2 below shows that c_n can be obtained in a distribution-free fashion and demonstrates the consistency of the proposed test (see Appendix F.15 for the proof).

Proposition E.2. *If the μ_j's are Lebesgue absolutely continuous, then under H_0, $\mathrm{RE}^2_{1:K,n}$ is a*

distribution-free statistic. Additionally, if $n_j/n \to \lambda_j \in (0,1)$ *such that* $\sum_{j=1}^{K} \lambda_j = 1$*, and the empirical distribution on* $\mathcal{H}_n^d$ *converges weakly to* $\mathcal{U}^d$*, then the above test which rejects* H_0 *when* $n\mathrm{RE}^2_{1:K,n} \geq c_n$*, is consistent.*

E.2 Comparison with competing rank-based approaches

This section is devoted to the comparisons between the approach proposed in this paper with those in other papers that define multivariate ranks using measure transportation. All the papers referenced in this section were posted before ours. A brief discussion on new papers in this area which were posted after the first version of our paper, has been added in Section 5 of the main paper.

We start off with [19], which in fact, was the main motivation behind our approach. Recall that we define population ranks using measure transportation from the population distribution (say μ) to the uniform distribution on $[0,1]^d$ (see Section 2 in the main paper). In [19], the authors suggest using a "special uniform distribution on the unit sphere" instead of our proposed uniform distribution on $[0,1]^d$. For our notion of population ranks, the following interesting property holds: given a "nice" measure $\mu = (\mu_1, \mu_2)$ with μ_1 and μ_2 supported on $\mathbb{R}^{d_1}$ and $\mathbb{R}^{d_2}$ respectively, suppose $F(\cdot),\ F_1(\cdot),\ F_2(\cdot)$ denote the corresponding population rank maps, then we must have $F = (F_1\ F_2)$ if and only if μ_1 and μ_2 are independent. In other words, the joint measure splitting into product of marginals is equivalent to the joint rank map splitting component-wise into marginal rank maps. This property is not true if we replace the uniform distribution on $[0,1]^d$ with the distribution proposed in [19]. This interesting observation allows us to obtain an elegant solution to the mutual independence testing problem, as can be seen, for example, in the proof of Lemma 3.1 in the main paper.

Two other notions of multivariate ranks have been proposed in [12, 16] based on what is called as the *semi-discrete optimal transportation problem*. Due to space constraints, we do not

describe this approach in detail here. However, the joint distribution of their proposed rank vectors is not distribution-free for fixed sample sizes. The theoretical results in [12] (including uniform convergence of rank maps) assume compactly supported distributions, whereas [16] proves a more general result under the condition that the population rank map is a homeomorphism — still a relatively strong assumption. Note that it is also very difficult to feasibly compute the notion of multivariate ranks proposed in [12, 16] beyond $d \geq 4$.

E.3 Quasi-Monte Carlo methods

Define $\mathfrak{F} := \{f : [0,1]^d \mapsto \mathbb{R} | \int_{[0,1]^d} f^2(\mathbf{x})\, d\mathbf{x} < \infty, f \in C^2([0,1]^d), f \text{ has bounded HKV}\}$; here HKV stands for *Hardy-Krause* variation, see [35] for detailed definitions. We begin with the following problem:

$$\textbf{(P)}: \qquad \text{Calculate/Approximate} \int_{[0,1]^d} f(\mathbf{x})\, d\mathbf{x}, \qquad \text{for } f \in \mathfrak{F}.$$

Problem **(P)** has several applications in engineering, astronomy, mathematical and computational finance, and has consequently been studied extensively across various disciplines (for a book length treatment see [42] and the extensive list of references therein). In most applications, difficulties in solving **(P)** arise from two main sources — (a) the antiderivative of $f(\cdot)$ may be hard to compute or it may not exist, and, (b) even a closed form expression for $f(\cdot)$ may not be available and the only information accessible to the analyst might be through evaluations of $f(\cdot)$ at certain query points of choice. Accordingly, the standard approach towards solving **(P)** is by choosing a set of query points $\mathbf{x}^{(n)} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, $\mathbf{x}_i \in [0,1]^d$ and combining the evaluations $\{f(\mathbf{x}_i)\}_{1 \leq i \leq n}$ by simple or weighted averaging. For the subsequent discussion, given any approximation $T_n\big(f; \mathbf{x}^{(n)}\big)$ to **(P)** based on the set of query points $\mathbf{x}^{(n)}$, we define the *approximation error* ($\epsilon_{n,f}$) as:

$$\epsilon_{n,f} := \left| \int_{[0,1]^d} f(\mathbf{x})\, d\mathbf{x} - T_n\big(f; \mathbf{x}^{(n)}\big) \right|.$$

One of the earliest approaches in literature was through the use of product numerical integration techniques. The principle idea is as follows: Suppose that $n = (m+1)^d$ for some $m \in \mathbb{N} \cup \{0\}$ and divide the $[0,1]$ interval along each of the coordinate axes into m subintervals (of equal width). Consider the d sets comprising of the endpoints from these intervals and form $\mathbf{x}^{(n)}$ by taking the Cartesian product of these d sets. The collection of evaluations $\{f(\mathbf{x}_i)\}_{1 \le i \le n}$ is then combined using a suitably chosen weighted averaging scheme. This idea forms the foundation of popular techniques such as the Simpson's rule, the Trapezoidal rule, etc. and leads to an *approximation error* of $O\big(n^{-2/d}\big)$ in general (see [3, 13]). Note that these *approximation errors* suffer from the "curse of dimensionality" and become less useful for even moderately large d.

A significant step in resolving this "curse of dimensionality" problem was achieved through the development of *Monte Carlo* methods (see [43, 21] and the references therein). The main idea here is to construct $\mathbf{x}^{(n)}$ using a random sample of n vectors according to the uniform distribution on $[0,1]^d$. The evaluations are then combined to form $T_n(f; \mathbf{x}^{(n)})$ as $n^{-1} \sum_{i=1}^n f(\mathbf{x}_i)$. In this case, $\epsilon_{n,f}$ (now a random variable) is $O_p\big(n^{-1/2}\big)$ (which follows from simple standard moment calculations). This dimension-free bound comes at a price. The error rates being random, there is no way to analyze the quality of approximation for a particular realization of $\mathbf{x}^{(n)}$. In certain sensitive practical problems, this can be a rather serious issue.

In subsequent research, the utility of *Monte Carlo* methods triggered this interesting idea: if a random sample corresponds to an average *approximation error* bound of $O_p\big(n^{-1/2}\big)$, then there must be deterministic sequences which lead to error bounds of at most $O\big(n^{-1/2}\big)$. This served as the main motivation behind the long line of work, now referred to as *Quasi-Monte Carlo* methods, which aims at finding fixed sequences which perform at least as well as *Monte Carlo* methods for solving **(P)** (uniformly in $f(\cdot)$). Before delving further into this area, let us bring in some relevant notation and results. Define $\mathbb{J} := \{\prod_{j=1}^d [0, u_j) : (u_1, \ldots, u_d) \in [0,1)^d\}$

and the *star discrepancy* of a set of vectors $\mathbf{x}^{(n)}$ as:

$$\mathcal{D}_n^*\big(\mathbf{x}^{(n)}\big) := \sup_{J\in\mathbb{J}} \left| \lambda_d(J) - \frac{1}{N}\cdot \#\{i : \mathbf{x}_i \in J\} \right|$$

where $\lambda_d(J)$ denotes the Lebesgue measure of the set J. The following is a classical result in Quasi-Monte Carlo literature.

Proposition E.3 (Koksma-Hlawka inequality, see [24])**.** *Suppose $f \in \mathfrak{F}$ and let $V(f)$ denote the HKV of $f(\cdot)$. Then, for any $\mathbf{x}^{(n)} := \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ with $\mathbf{x}_i \in [0,1)^d$, we have:*

$$\left| \int_{[0,1]^d} f(\mathbf{x})\, d\mathbf{x} - \frac{1}{N}\sum_{i=1}^{n} f(\mathbf{x}_i) \right| \le V(f)\mathcal{D}_n^*\big(\mathbf{x}^{(n)}\big).$$

Proposition E.3 above implies that an upper bound to the star discrepancy of a sequence yields an upper bound to the *approximation error*. As a result, constructing sequences with small values of the star discrepancy has become a popular approach to addressing **(P)** over the past 50 years or so. Some examples include *Kronecker sequences*, *Halton sequences* and digital sequences of *Niederreiter type* (see [20, 31, 39]). Let us describe one of them, namely the *Halton sequence*.

Halton sequences: Given n, d, for $1 \le k \le n$, $1 \le j \le d$, form $x(k,j)$ as follows: let p_i denote the ith prime and express k in base p_i. Invert this p_i-ary expansion and place it after the decimal point. For example, in the case of $x(6,1)$, note that $6 = (110)_2$, which makes $x(6,1) = (0.011)_2 = 3/8$. Finally, the *Halton sequence* is formed by setting $\mathbf{x}^{(n)} := \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ where $\mathbf{x}_i := (x(i,1), \ldots, x(i,d)) \in [0,1]^d$. The *star discrepancy* for these sequences is $O\big(\log^d(n)/n\big)$; see e.g., [25, 26].

Corollary E.1. *Let $\mathbb{P}_n$ denote the empirical measure on a d-dimensional quasi-Monte Carlo sequence (see the examples above) of n vectors. Then $\mathbb{P}_n \xrightarrow{w} \mathcal{U}^d$. A similar result holds for the regular griding scheme used in product numerical integration or the random sampling scheme used in Monte Carlo methods.*

The above discussion is a very brief overview of this active research area. For the purposes of this paper it is perhaps instructive to filter out some key desirable properties of quasi-Monte Carlo sequences with low star-discrepancy (e.g., Halton sequences) which will be useful to us. Although we restricted ourselves to the function class $\mathfrak{F}$ in this section, it is possible to consider

Properties	Regular grid	Random sampling	Halton sequences
Sequence is deterministic	Yes	No	Yes
Can be constructed for all n, d	No, only when n is a perfect power of d	Yes	Yes
Rate of star discrepancy	$n^{-2/d}$	$n^{-1/2}$	$\log^d(n)/n$
For fixed d, whole sequence must be recomputed if n increases by 1	Yes	No, only one vector needs to be computed	No, only one vector needs to be computed

Table E.1: Properties of sequences discussed above. The most desirable properties are highlighted in blue.

more general function classes. However these technical details play no role in this paper, and hence we do not elaborate on these issues.

F Proofs

This section contains proofs of our main results.

F.1 Proof of Proposition 2.2

Proof. **(i)** This part of the proof is verbatim identical to [19, Proposition 1.6.1].

(ii) First, recall the definition of $\widehat{\sigma}_n$ from (2.5) in the main paper. Next, note that $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ are exchangeable (as they are i.i.d.). Moreover, $\mathbf{X}_i$'s are absolutely continuous. This implies that, given any two permutations σ_1, σ_2 in S_n, we have:

$$\sum_{i=1}^{n} \|\mathbf{X}_i - \mathbf{h}_{\sigma_1(i)}\|^2 \neq \sum_{i=1}^{n} \|\mathbf{X}_i - \mathbf{h}_{\sigma_2(i)}\|^2 \qquad \mu\text{-a.e.} \tag{F.1}$$

Moreover, the following set of $n!$ random variables, $\left\{\sum_{i=1}^{n}\|\mathbf{X}_{\sigma(i)}-\mathbf{h}_i\|^2\right\}_{\sigma\in S_n}$ forms an exchangeable collection. Coupled with (F.1) the above observation yields that $\mathbb{P}(\widehat{\sigma}_n=\sigma)=(n!)^{-1}$ for all $\sigma\in S_n$. This completes the proof.

(iii) The independence between the multivariate ranks and the multivariate order statistics is now a direct consequence of the observation that the order statistics are complete sufficient (from part **(i)**), the multivariate ranks are ancillary for μ (as their distribution is free of μ from part **(ii)**), followed by an application of Basu's Theorem. □

F.2 Proof of Theorem 2.1

Proof. This proof requires a number of existing results from convex analysis which we present in Appendix H for the convenience of the reader.

Recall that μ_n is defined as the empirical measure on $\mathcal{D}_n=\{\mathbf{X}_1,\ldots,\mathbf{X}_n\}$ (see (2.3) in the main paper) where $\mathbf{X}_1,\ldots,\mathbf{X}_n\overset{i.i.d.}{\sim}\mu$. Also let all the random variables, i.e., $\mathbf{X}_i$'s be defined on the probability space $(\widetilde{\Omega},\mathcal{A},\mathbb{P})$. We know that $\mu_n\xrightarrow{w}\mu$ a.e. (see [53, Theorem 1]). In other words, there exists $\Omega\subset\widetilde{\Omega}$, such that $\mathbb{P}(\Omega)=1$ and, for all $\omega\in\Omega$, $\mu_n(\omega)\xrightarrow{w}\mu$.

Fix $\omega\in\Omega$. Now consider the sequence of random measures $(\text{identity},\widehat{\mathbf{R}}_n)\#\mu_n(\omega)$; the randomness comes the random draw according to $\mu_n(\omega)$. As $\widehat{\mathbf{R}}_n(\omega)$ takes values in a compact set and $\mu_n(\omega)$ converges weakly, we have that $(\text{identity},\widehat{\mathbf{R}}_n)\#\mu_n(\omega)$ is asymptotically tight. Consequently, by Prokhorov's theorem, every subsequence of $(\text{identity},\widehat{\mathbf{R}}_n)\#\mu_n(\omega)$ has a further subsequence, say $(\text{identity},\widehat{\mathbf{R}}_{n_k})\#\mu_{n_k}(\omega)$ (by a relabeling if necessary) such that $(\text{identity},\widehat{\mathbf{R}}_{n_k})\#\mu_{n_k}(\omega)$ converges weakly on $\mathbb{R}^d\times\mathbb{R}^d$. Let us call this limit $\gamma(\omega)$ (which could also depend on the subsequence $\{n_k\}_{k\geq 1}$). Next, we will show that each of these subsequences have the same weak limit (which for now may depend on ω).

Let $\Gamma(\mu,\mathcal{U}^d)$ be the family of probability distributions on $\mathbb{R}^d\times\mathbb{R}^d$ that have marginals (on the first and the last d-coordinates) μ and $\mathcal{U}^d$ respectively. Also, note that, by (2.5), (2.6) in the main paper, and Definition H.1, $(\text{identity},\widehat{\mathbf{R}}_{n_k})\#\mu_{n_k}(\omega)$ has cyclically monotone support. As $\mu_{n_k}(\omega)\xrightarrow{w}\mu$ and $\widehat{\mathbf{R}}_{n_k}\#\mu_{n_k}(\omega)$ (which is simply the empirical measure on $\mathcal{H}^d_{n_k}$) converges weakly to $\mathcal{U}^d$ (by assumption), we can conclude that $\gamma(\omega)$ has cyclically monotone support and $\gamma(\omega)\in\Gamma(\mu,\mathcal{U}^d)$ (by [34, Lemma 9]). Moreover, by [34, Corollary 14] and the fact that $\mu\in\mathcal{P}_{ac}(\mathbb{R}^d)$, there exists only one measure with cyclically monotone support in $\Gamma(\mu,\mathcal{U}^d)$. Therefore, irrespective of the subsequence $\{n_k\}_{k\geq 1}$, $(\text{identity},\widehat{\mathbf{R}}_{n_k})\#\mu_{n_k}(\omega)$ converges weakly to the same limit.

The above also shows that the weak limit is the same for all $\omega\in\Omega$; let us call it γ. Finally, by [34, Proposition 10] and the definition of $\mathbf{R}(\cdot)$, we get that $\gamma=(\text{identity},\mathbf{R})\#\mu$. Therefore, we have proved that

$$(\text{identity},\widehat{\mathbf{R}}_n)\#\mu_n(\omega)\xrightarrow{w}(\text{identity},\mathbf{R})\#\mu\qquad\text{for all }\ \omega\in\Omega. \tag{F.2}$$

Now, let $\mathbf{M}_n$ be sampled according to the measure $\mu_n(\omega)$ and $\mathbf{M}$ be sampled according to the measure μ. Then (F.2) can be restated as:

$$(\mathbf{M}_n, \widehat{\mathbf{R}}_n(\mathbf{M}_n)) \xrightarrow{w} (\mathbf{M}, \mathbf{R}(M)) \qquad \text{for all } \omega \in \Omega. \tag{F.3}$$

Let $g : \mathbb{R}^d \times \mathbb{R}^d \to [0, \infty)$ be defined as $g(\mathbf{x}, \mathbf{y}) := \|\mathbf{y} - \mathbf{R}(x)\|$. Note that by Alexandroff theorem (see e.g., [2]), $\mathbf{R}(\cdot)$ is continuous Lebesgue a.e., and consequently μ-a.e. (by the absolute continuity of μ). Therefore the function $g(\cdot)$ is discontinuous on a set which has measure 0 with respect to $(\text{identity}, \mathbf{R})\#\mu$. Consequently, by applying the continuous mapping theorem with $g(\cdot)$ on (F.3), we get, for all $\omega \in \Omega$,

$$g(\mathbf{M}_n, \widehat{\mathbf{R}}_n(\mathbf{M}_n)) = \|\widehat{\mathbf{R}}_n(\mathbf{M}_n) - \mathbf{R}(\mathbf{M}_n)\| \xrightarrow{w} g(M, R(M)) = 0.$$

Finally, as $\widehat{\mathbf{R}}_n$ and $\mathbf{R}$ are uniformly bounded, the dominated convergence theorem implies,

$$\int \|\widehat{\mathbf{R}}_n(\cdot) - \mathbf{R}(\cdot)\| \, d\mu_n(\omega) = \frac{1}{n} \sum_{i=1}^{n} \|\widehat{\mathbf{R}}_n(\mathbf{X}_i) - \mathbf{R}(\mathbf{X}_i)\|(\omega) \longrightarrow 0 \quad \text{for all } \omega \in \Omega.$$

This completes the proof. □

F.3 Proof of Proposition 3.1

Proof. For the solution of this problem, as we are interested in limits under $|t|/|s| \to c$. Thus, let us assume without loss of generality that $|t|/|s| \in (c/2, 2c)$. Note that, by a first order Taylor series expansion of the exponential function (as $G_1(Z_1)$ and $G_2(Z_2)$ are uniformly bounded),

$$\begin{aligned} F(t,s) &:= \mathbb{E}\left[\exp(itG_1(Z_1) + isG_2(Z_2))\right] \\ &= 1 + it\mathbb{E}[G_1(Z_1)] + is\mathbb{E}[G_2(Z_2)] - \frac{t^2}{2}\mathbb{E}[G_1(Z_1)^2] - \frac{s^2}{2}\mathbb{E}[G_2(Z_2)^2] - st\mathbb{E}[G_1(Z_1)G_2(Z_2)] \\ &\quad + \mathcal{O}(\max\{|s|^3, |t|^3\}). \end{aligned} \tag{F.4}$$

Now plugging in $t = 0$, followed by $s = 0$, alternatively in (F.4) and multiplying the results, we get:

$$\begin{aligned} F(t,0)F(0,s) &= \mathbb{E}\left[\exp(itG_1(Z_1)\right] \mathbb{E}\left[isG_2(Z_2)\right] \\ &= 1 + it\mathbb{E}[G_1(Z_1)] + is\mathbb{E}[G_2(Z_2)] - \frac{t^2}{2}\mathbb{E}[G_1(Z_1)^2] - \frac{s^2}{2}\mathbb{E}[G_2(Z_2)^2] \\ &\quad - st\mathbb{E}[G_1(Z_1)]\mathbb{E}[G_2(Z_2)] + \mathcal{O}(\max\{|s|^3, |t|^3\}). \end{aligned}$$

This implies, following the notation from the problem statement,

$$f_{Z_1,Z_2}(t,s) = F(t,s) - F(t,0)F(0,s)$$

$$= st\mathrm{Cov}(G_1(Z_1), G_2(Z_2)) + \mathcal{O}(\max\{|s|^3, |t|^3\}). \tag{F.5}$$

In the same vein as (F.5), we further get the following estimates:

$$f_{Z_1,Z_1}(t,t) = t^2\mathrm{Var}[G_1(Z_1)] + \mathcal{O}(|t|^3)\ ,\ f_{Z_2,Z_2}(s,s) = s^2\mathrm{Var}[G_2(Z_2)] + \mathcal{O}(|s|^3). \tag{F.6}$$

By combining these observations from (F.5) and (F.6), we get:

$$\begin{aligned}
&\lim_{t,s\to 0, |t|/|s|\to c} \frac{|f_{Z_1,Z_2}(t,s)|^2}{|f_{Z_1,Z_1}(t,t)||f_{Z_2,Z_2}(s,s)|} \\
&= \lim_{t,s\to 0, |t|/|s|\to c} \frac{(st\mathrm{Cov}(G_1(Z_1), G_2(Z_2)) + \mathcal{O}(\max\{|t|^3, |s|^3\}))^2}{(t^2\mathrm{Var}[G_1(Z_1)] + \mathcal{O}(|t|^3))(s^2\mathrm{Var}[G_1(Z_1)] + \mathcal{O}(|s|^3))} \\
&= \rho^2(G_1(Z_1), G_2(Z_2)).
\end{aligned}$$

This completes the proof. □

F.4 Proof of Lemma 3.1

Proof. (a) This follows directly from the properties of distance covariance in [50, Remark 3] by noting that rank distance covariance is just the usual distance covariance between the multivariate ranks $\mathbf{R}_1(\mathbf{Z}_1)$ and $\mathbf{R}_2(\mathbf{Z}_2)$.

(b) (if part). Note that if $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are independent, then so are $\mathbf{R}_1(\mathbf{Z}_1)$ and $\mathbf{R}_2(\mathbf{Z}_2)$. Then, (3.2) in the main paper clearly implies $\mathrm{RdCov}^2(\mathbf{Z}_1, \mathbf{Z}_2) = 0$.

(only if part). By [50] (or (3.2) in the main paper coupled with continuity of characteristic functions), $\mathrm{RdCov}^2(\mathbf{Z}_1, \mathbf{Z}_2) = 0$ implies $\mathbf{R}_1(\mathbf{Z}_1)$ and $\mathbf{R}_2(\mathbf{Z}_2)$ are independent. Now note that, by Proposition 2.1 in the main paper, there exists maps $\mathbf{Q}_1 : [0,1]^{d_1} \to \mathbb{R}^{d_1}$ and $\mathbf{Q}_2 : [0,1]^{d_2} \to \mathbb{R}^{d_2}$ such that

$$\mathbf{Q}_1(\mathbf{R}_1(\mathbf{Z}_1)) = \mathbf{Z}_1 \ \ a.e.\ \mu_1 \qquad \text{and} \qquad \mathbf{Q}_2(\mathbf{R}_2(\mathbf{Z}_2)) = \mathbf{Z}_2 \ \ a.e.\ \mu_2.$$

By the above display, $(\mathbf{Z}_1, \mathbf{Z}_2) \stackrel{d}{=} (\mathbf{Q}_1(\mathbf{R}_1(\mathbf{Z}_1)), \mathbf{Q}_2(\mathbf{R}_2(\mathbf{Z}_2)))$. As $\mathbf{R}_1(\mathbf{Z}_1)$ and $\mathbf{R}_2(\mathbf{Z}_2)$ are independent, so are $\mathbf{Q}_1(\mathbf{R}_1(\mathbf{Z}_1))$ and $\mathbf{Q}_2(\mathbf{R}_2(\mathbf{Z}_2))$, and consequently, so are $\mathbf{Z}_1$ and $\mathbf{Z}_2$.

(c) From [50, Theorem 4], we know that $\mathrm{RdCov}(\mathbf{Z}_1, \mathbf{Z}_1) \geq 0$ with equality holding if and only if $\mathbf{R}_1(\mathbf{Z}_1)$ has a degenerate distribution. However, $\mathbf{R}_1(\mathbf{Z}_1)$ has a non-degenerate d-dimensional uniform distribution. So, $\mathrm{RdCov}(\mathbf{Z}_1, \mathbf{Z}_1) > 0$.

(d) Set $\mathbf{Y}_1 := \mathbf{a}_1 + b\mathbf{Z}_1$ where $\mathbf{a}_1 \in \mathbb{R}^{d_1}$ and $b > 0$. By Proposition 2.1 in the main paper, we know that the (unique) rank map is the gradient of a convex function. Let the convex function corresponding to $\mathbf{Z}_1$ be $\phi_1(\cdot)$. This implies that $b\phi_1(b^{-1}(\mathbf{x} - \mathbf{a}_1))$ is also a convex function and its gradient is given by $\mathbf{R}_1(b^{-1}(\mathbf{x} - \mathbf{a}_1))$. Set $\widetilde{\mathbf{R}}(\mathbf{y}_1) = \mathbf{R}_1(b^{-1}(\mathbf{y}_1 - \mathbf{a}_1))$ and note that

$$\widetilde{\mathbf{R}}(\mathbf{Y}_1) = \mathbf{R}_1(b^{-1}(\mathbf{a}_1 + b\mathbf{Z}_1 - \mathbf{a}_1)) = \mathbf{R}_1(\mathbf{Z}_1) \stackrel{d}{=} \mathcal{U}^{d_1}. \tag{F.7}$$

Therefore $\widetilde{\mathbf{R}}(\cdot)$ pushes the measure induced by $\mathbf{Y}_1$ to the d-dimensional uniform distribution, and it is the gradient of a convex function. As a result, Proposition 2.1 implies $\widetilde{\mathbf{R}}(\cdot)$ is the population rank map corresponding to $\mathbf{Y}_1$ is $\widetilde{\mathbf{R}}(\cdot)$.

The same argument as in (F.7) also yields that $\overline{\mathbf{R}}_2(\mathbf{Y}_2) = R_2(\mathbf{Z}_2)$ where $\mathbf{Y}_2 := \mathbf{a}_2 + b\mathbf{Z}_2$ and $\overline{\mathbf{R}}(\cdot)$ denotes the population rank map corresponding to $\mathbf{Y}_2$. Therefore, $\mathrm{RdCov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathrm{RdCov}(\mathbf{Z}_1, \mathbf{Z}_2)$.

(e) Let $\mathbf{R}_1^n$ and $\mathbf{R}_2^n$ denote the rank maps corresponding to $\mathbf{Z}_1^n$ and $\mathbf{Z}_2^n$ respectively. By repeating the same argument as in the proof of Theorem 2.1 in the main paper, we get:

$$(\mathbf{Z}_1^n, \mathbf{R}_1^n(\mathbf{Z}_1^n)) \xrightarrow{w} (\mathbf{Z}_1, \mathbf{R}_1(\mathbf{Z}_1)) \qquad \text{and} \qquad (\mathbf{Z}_2^n, \mathbf{R}_2^n(\mathbf{Z}_2^n)) \xrightarrow{w} (\mathbf{Z}_2, \mathbf{R}_2(\mathbf{Z}_2))$$

which then by the continuous mapping theorem applied to the function $g(\mathbf{y}, \mathbf{x}) := \|\mathbf{y} - \mathbf{R}_1(\mathbf{x})\|$ (equivalently, $g(\mathbf{y}, \mathbf{x}) = \|\mathbf{y} - \mathbf{R}_2(\mathbf{x})\|$), as in the proof of Theorem 2.1, implies,

$$\|\mathbf{R}_1^n(\mathbf{Z}_1^n) - \mathbf{R}_1(\mathbf{Z}_1^n)\| \xrightarrow{P} 0 \qquad \text{and} \qquad \|\mathbf{R}_2^n(\mathbf{Z}_2^n) - \mathbf{R}_2(\mathbf{Z}_2^n)\| \xrightarrow{P} 0. \tag{F.8}$$

Let $(\mathbf{Z}_1^{1,n}, \mathbf{Z}_2^{1,n}), (\mathbf{Z}_1^{2,n}, \mathbf{Z}_2^{2,n}), (\mathbf{Z}_1^{3,n}, \mathbf{Z}_2^{3,n})$ be i.i.d. according to the same distribution as $(\mathbf{Z}_1^n, \mathbf{Z}_2^n)$ and let $(\mathbf{Z}_1^1, \mathbf{Z}_2^1), (\mathbf{Z}_1^2, \mathbf{Z}_2^2), (\mathbf{Z}_1^3, \mathbf{Z}_2^3)$ be i.i.d. copies of $(\mathbf{Z}_1, \mathbf{Z}_2)$. Next, observe that, by the triangle inequality,

$$\begin{aligned}
&\Big| \|\mathbf{R}_1^n(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1^n(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2^n(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2^n(\mathbf{Z}_2^{2,n})\| - \|\mathbf{R}_1(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2(\mathbf{Z}_2^{1,n})\| \Big| \\
&\leq \|\mathbf{R}_1^n(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1(\mathbf{Z}_1^{1,n})\| \|\mathbf{R}_2^n(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2^n(\mathbf{Z}_2^{2,n})\| + \|\mathbf{R}_1^n(\mathbf{Z}_1^{2,n}) - \mathbf{R}_1(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2^n(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2^n(\mathbf{Z}_2^{2,n})\| \\
&\qquad + \|\mathbf{R}_1(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2^n(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2(\mathbf{Z}_2^{1,n})\| \\
&\qquad + \|\mathbf{R}_1(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2^n(\mathbf{Z}_2^{2,n}) - \mathbf{R}_2(\mathbf{Z}_2^{2,n})\| \xrightarrow{P} 0.
\end{aligned}$$

Finally note that, by using the continuous mapping theorem on the joint weak convergence of $(\mathbf{Z}_1^n, \mathbf{Z}_2^n)$ to $(\mathbf{Z}_1, \mathbf{Z}_2)$, we further get:

$$\|\mathbf{R}_1(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2(\mathbf{Z}_2^{2,n})\| \xrightarrow{w} \|\mathbf{R}_1(\mathbf{Z}_1^1) - \mathbf{R}_1(\mathbf{Z}_1^2)\| \|\mathbf{R}_1(\mathbf{Z}_2^1) - \mathbf{R}_1(\mathbf{Z}_2^2)\|.$$

Next, by applying the dominated convergence theorem, we get:

$$\begin{aligned}
&\mathbb{E}\big[\|\mathbf{R}_1(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2(\mathbf{Z}_2^{2,n})\|\big] \\
&\xrightarrow{n\to\infty} \mathbb{E}\big[\|\mathbf{R}_1(\mathbf{Z}_1^1) - \mathbf{R}_1(\mathbf{Z}_1^2)\| \|\mathbf{R}_1(\mathbf{Z}_2^1) - \mathbf{R}_1(\mathbf{Z}_2^2)\|\big].
\end{aligned} \tag{F.9}$$

Combining (F.9), (F.8) with the dominated convergence theorem yields:

$$\begin{aligned}
&\mathbb{E}\big[\|\mathbf{R}_1^n(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1^n(\mathbf{Z}_1^{2,n})\| \|\mathbf{R}_2^n(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2^n(\mathbf{Z}_2^{2,n})\|\big] \\
&\xrightarrow{n\to\infty} \mathbb{E}\big[\|\mathbf{R}_1(\mathbf{Z}_1^1) - \mathbf{R}_1(\mathbf{Z}_1^2)\| \|\mathbf{R}_1(\mathbf{Z}_2^1) - \mathbf{R}_1(\mathbf{Z}_2^2)\|\big].
\end{aligned} \tag{F.10}$$

By using the same arguments as above, we can similarly show the following:

$$\begin{aligned}&\mathbb{E}\big[\|\mathbf{R}_1^n(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1^n(\mathbf{Z}_1^{2,n})\|\big]\mathbb{E}\big[\|\mathbf{R}_2^n(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2^n(\mathbf{Z}_2^{2,n})\|\big]\\ &\overset{n\to\infty}{\longrightarrow} \mathbb{E}\big[\|\mathbf{R}_1(\mathbf{Z}_1^1) - \mathbf{R}_1(\mathbf{Z}_1^2)\|\big]\mathbb{E}\big[\|\mathbf{R}_1(\mathbf{Z}_2^1) - \mathbf{R}_1(\mathbf{Z}_2^2)\|\big]\end{aligned} \tag{F.11}$$

and

$$\begin{aligned}&\mathbb{E}\big[\|\mathbf{R}_1^n(\mathbf{Z}_1^{1,n}) - \mathbf{R}_1^n(\mathbf{Z}_1^{2,n})\|\|\mathbf{R}_2^n(\mathbf{Z}_2^{1,n}) - \mathbf{R}_2^n(\mathbf{Z}_2^{3,n})\|\big]\\ &\overset{n\to\infty}{\longrightarrow} \mathbb{E}\big[\|\mathbf{R}_1(\mathbf{Z}_1^1) - \mathbf{R}_1(\mathbf{Z}_1^2)\|\|\mathbf{R}_1(\mathbf{Z}_2^1) - \mathbf{R}_1(\mathbf{Z}_2^3)\|\big].\end{aligned} \tag{F.12}$$

Combining (F.10), (F.11), (F.12) and Lemma 3.1 (part (a)) in the main paper, we finally get:

$$\mathrm{RdCov}^2(\mathbf{Z}_1^n, \mathbf{Z}_2^n) \overset{n\to\infty}{\longrightarrow} \mathrm{RdCov}^2(\mathbf{Z}_1, \mathbf{Z}_2)$$

which completes the proof. □

F.5 Proof of Lemma 3.2

Proof. (a) This is a direct consequence of [4, Lemmas 2.2 and 2.3].

(b) (if part). Assuming $\mathbf{Z}_1 \overset{d}{=} \mathbf{Z}_2$, we have $\mathbb{P}(\mathbf{a}^\top\mathbf{Z}_1 \le t) = \mathbb{P}(\mathbf{a}^\top\mathbf{Z}_2 \le t)$ for all $\mathbf{a} \in \mathcal{S}^{d-1}$ and $t \in \mathbb{R}$. Therefore, by (3.5) in the main paper, $\mathrm{RE}_\lambda^2(\mathbf{Z}_1, \mathbf{Z}_2) = 0$.

(only if part). By [4, Theorem 2.1], if $\mathrm{RE}_\lambda^2(\mathbf{Z}_1, \mathbf{Z}_2) = 0$, then we have:

$$\mathbf{R}_\lambda(\mathbf{Z}_1) \overset{d}{=} \mathbf{R}_\lambda(\mathbf{Z}_2). \tag{F.13}$$

Next, by Proposition 2.1 in the main paper, there exists $\mathbf{Q}_\lambda : \mathbb{R}^d \to \mathbb{R}^d$ such that

$$\mathbf{Q}_\lambda(\mathbf{R}_\lambda(\mathbf{Z}_1)) = \mathbf{Z}_1 \quad \text{a.e. } \mu_1 \quad \text{and} \quad \mathbf{Q}_\lambda(\mathbf{R}_\lambda(\mathbf{Z}_2)) = \mathbf{Z}_2 \quad \text{a.e. } \mu_2. \tag{F.14}$$

Finally, (F.14) combined with (F.13) yields $\mathbf{Z}_1 \overset{d}{=} \mathbf{Z}_2$ and completes the proof.

(c) The proof is verbatim similar to that of Lemma 3.1 (part (d)) in Appendix F.4.

(d) Note that, by part (a), $\mathrm{RE}_\lambda^2(\mathbf{Z}_1, \mathbf{Z}_2)$ can be written as expectations of Euclidean distances between bounded random vectors. Therefore, the proof is exactly similar to that of Lemma 3.1 (part (e)) in Appendix F.4. We leave the details to the reader. □

F.6 Proof of Lemma 4.1

Proof. Note that RdCov_n^2, as defined in (4.1) in the main paper, is a function of $(\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_1), \ldots, \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_n), \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_1), \ldots, \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_n))$. Under H_0, the distribution of the above vec-

tor further splits into the product of the marginal distributions of $(\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_1), \ldots, \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_n))$ and $(\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_1), \ldots, \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_n))$. By Proposition 2.2 (part (ii)) in the main paper, each of the marginals are distribution-free, i.e., their distribution does not depend on $\mu_{\mathbf{X}}$ and $\mu_{\mathbf{Y}}$. In fact, they are distributed uniformly over all $n!$ permutations of each of the fixed grids $\mathcal{H}_n^{d_1}$ and $\mathcal{H}_n^{d_2}$ respectively. This results in the distribution-free property of the statistic RdCov_n^2 (under H_0). □

F.7 Proof of Lemma 4.2

Proof. Let us first prove (4.2) in the main paper. Suppose that $(X_1, Y_1), (X_2, Y_2)$ and (X_3, Y_3) are random samples drawn from the joint distribution of (X, Y). Let $W_i := F^X(X_i)$ and $Z_i := F^Y(Y_i)$ for $i = 1, 2, 3$. Further, we will use $F^{W,Z}(\cdot)$, $F^W(\cdot)$ and $F^Z(\cdot)$ to denote the joint and marginal DFs of W and Z respectively. Note that,

$$\mathrm{RdCov}^2(X, Y) = \mathbb{E}[|W_1 - W_2||Z_1 - Z_2|] + \mathbb{E}|W_1 - W_2|\mathbb{E}|Z_1 - Z_2| - 2\mathbb{E}|W_1 - W_2||Z_1 - Z_3|. \quad \text{(F.15)}$$

Further, we can write the following simple algebraic identity:

$$|W_1 - W_2| = \int_{-\infty}^{\infty} \left[\mathbf{1}(W_1 \le u \le W_2) + \mathbf{1}(W_2 \le u \le W_1)\right] du. \quad \text{(F.16)}$$

We can write a similar result for $|Z_1 - Z_2|$ (as in (F.16)), which, on multiplying with the right hand side of (F.16), yields:

$$\begin{aligned} &|W_1 - W_2||Z_1 - Z_2| \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \Big[\mathbf{1}(W_1 \le u \le W_2)\mathbf{1}(Z_1 \le v \le Z_2) + \mathbf{1}(W_1 \le u \le W_2)\mathbf{1}(Z_2 \le v \le Z_1) \\ &\quad + \mathbf{1}(W_2 \le u \le W_1)\mathbf{1}(Z_1 \le v \le Z_2) + \mathbf{1}(W_2 \le u \le W_1)\mathbf{1}(Z_2 \le v \le Z_1)\, du\, dv\Big] \end{aligned} \quad \text{(F.17)}$$

Next, by applying Fubini's theorem on (F.17), we get:

$$\begin{aligned} \mathbb{E}\left[|W_1 - W_2||Z_1 - Z_2|\right] = 2\int_{-\infty}^{\infty} \Big[&F^{W,Z}(u,v) + 2(F^{W,Z}(u,v))^2 - 2F^W(u)F^{W,Z}(u,v) \\ &- 2F^{W,Z}(u,v)F^Z(v) + F^W(u)F^Z(v)\Big]\, du\, dv. \end{aligned} \quad \text{(F.18)}$$

Similar calculations also result in the following:

$$\begin{aligned} \mathbb{E}|W_1 - W_2|\,\mathbb{E}|Z_1 - Z_2| = 4\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \Big[&F^W(u)F^Z(v) - (F^W(u))^2F^Z(v) \\ &- F^W(u)(F^Z(v))^2 + (F^W(u)F^Z(v))^2\Big]\, du\, dv \end{aligned} \quad \text{(F.19)}$$

and,

$$\mathbb{E}|W_1 - W_2||Z_1 - Z_3| = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\Big[3F^W(u)F^Z(v) + F^{W,Z}(u,v) + 4F^{W,Z}(u,v)F^W(u)F^Z(v) \\ - 2F^W(u)F^{W,Z}(u,v) - 2F^{W,Z}(u,v)F^Z(v) - 2(F^W(u))^2F^Z(v) \\ - 2F^Z(u)(F^W(v))^2\Big]\,du\,dv. \tag{F.20}$$

Therefore, by combining (F.15), (F.18), (F.19) and (F.20), we get:

$$\begin{aligned}\mathrm{RdCov}^2(X,Y) &= 4\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\Big[(F^{W,Z}(u,v))^2 - 2F^W(u)F^Z(v)F^{W,Z}(u,v) + (F^W(u)F^Z(v))^2\Big]\,du\,dv \\ &= 4\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\Big(F^{W,Z}(u,v) - F^W(u)F^Z(v)\Big)^2\,du\,dv.\end{aligned}$$

Next, note that $F^{W,Z}(u,v) = F^{X,Y}\big((F^X(u))^{-1},(F^Y(v))^{-1}\big)$, $F^W(u) = u$ and $F^Z(v) = v$. Using this observation, coupled with a standard change of variable formula for Riemann integrals, we get:

$$\frac{1}{4}\mathrm{RdCov}^2(X,Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\big(F^{X,Y}(x,y) - F^X(x)F^Y(y)\big)^2\,dF^X(x)\,dF^Y(y).$$

This completes the proof of (4.2) in the main paper.

In order to prove (4.3) in the main paper, let us start with some notation:

$$T_1 := \frac{1}{n^4}\sum_{i,j,k,l}\mathbf{1}(X_l < X_j, Y_k < Y_i)\ ,\ T_2 := \frac{1}{n^5}\sum_{i,j,k,l,m}\mathbf{1}(X_l < X_j, Y_k < Y_m, Y_i < Y_m)$$

$$T_3 := \frac{1}{n^5}\sum_{i,j,k,l,m}\mathbf{1}(Y_i < Y_j, X_k < X_m, X_l < X_m)\ ,\ T_4 := \frac{1}{n^3}\sum_{i,j,k}\mathbf{1}(X_k < X_i, Y_k < Y_j),$$

$$T_5 := \frac{1}{n^4}\sum_{i,j,k,l}\mathbf{1}(X_l < X_j)\mathbf{1}(Y_k < Y_i, Y_l < Y_i),$$

$$T_6 := \frac{1}{n^4}\sum_{i,j,k,l}\mathbf{1}(X_k < X_j, X_l < X_j)\mathbf{1}(Y_k < Y_i),$$

$$T_7 := \frac{1}{n^5}\sum_{i,j,k,l,m}\mathbf{1}(X_k < X_j, X_l < X_j)\mathbf{1}(Y_k < Y_i, Y_m < Y_i),$$

$$T_8 := \frac{1}{n^4}\sum_{i,j,k,l}\mathbf{1}(X_k < X_j, X_l < X_j)\mathbf{1}(Y_k < Y_i, Y_l < Y_i),$$

$$T_9 := \frac{1}{n^6} \sum_{i,j,k,l,m,p} \mathbf{1}(X_k < X_j, X_l < X_j)\mathbf{1}(Y_m < Y_i, Y_p < Y_i).$$

Next, note that:

$$\begin{aligned} S_1 &:= \frac{1}{n^2} \sum_{k,l} \left|\widehat{R}_n^X(X_k) - \widehat{R}_n^X(X_l)\right|\left|\widehat{R}_n^Y(Y_k) - \widehat{R}_n^Y(Y_l)\right| \\ &= \frac{1}{n^4} \sum_{k,l,i,j} \left[\mathbf{1}(X_k < X_j < X_l) + \mathbf{1}(X_l < X_j < X_k)\right]\left[\mathbf{1}(Y_k < Y_i < Y_l) + \mathbf{1}(Y_l < Y_i < Y_k)\right] \\ &= 4T_1 - 4T_2 - 4T_3 + 4T_9. \end{aligned} \tag{F.21}$$

Similar calculations reveal that,

$$\begin{aligned} S_2 &:= \left(\frac{1}{n^2} \sum_{k,l=1}^{n} \left|\widehat{R}_n^X(X_k) - \widehat{R}_n^X(X_l)\right|\right) \times \left(\frac{1}{n^2} \sum_{k,l=1}^{n} \left|\widehat{R}_n^Y(Y_k) - \widehat{R}_n^Y(Y_l)\right|\right) \\ &= 2T_1 + 2T_4 - 4T_5 - 4T_6 + 4T_8, \end{aligned} \tag{F.22}$$

and

$$\begin{aligned} S_3 &:= \frac{1}{n^3} \sum_{k,l,m=1}^{n} \left|\widehat{R}_n^X(X_k) - \widehat{R}_n^X(X_l)\right|\left|\widehat{R}_n^Y(Y_k) - \widehat{R}_n^Y(Y_m)\right| \\ &= 3T_1 - 2T_2 - 2T_3 + T_4 - 2T_5 - 2T_6 + 4T_7. \end{aligned} \tag{F.23}$$

Recall that $\mathrm{RdCov}_n^2 = S_1 + S_2 - 2S_3$. Therefore, by (F.21), (F.22) and (F.23), we get:

$$\begin{aligned} \mathrm{RdCov}_n^2 &= 4(T_7 - 2T_8 + T_9) \\ &= \frac{4}{n^2} \sum_{i,j} \left(F_n^{X,Y}(X_i, Y_j) - F_n^X(X_i)F_n^Y(Y_j)\right)^2 \\ &= 4 \int\int \left(F_n^{X,Y}(x,y) - F_n^X(x)F_n^Y(y)\right)^2 dF_n^X(x)\, dF_n^Y(y). \end{aligned}$$

This completes the proof of (4.3) in the main paper. □

F.8 Proof of Theorem 4.1

Proof. Let us start with some notation. Let $\mathcal{H}_n^{d_1} = \{\mathbf{h}_1^{d_1}, \mathbf{h}_2^{d_1}, \ldots, \mathbf{h}_n^{d_1}\}$ and $\mathcal{H}_n^{d_2} = \{\mathbf{h}_1^{d_2}, \mathbf{h}_2^{d_2}, \ldots, \mathbf{h}_n^{d_2}\}$. Next, define:

$$w(\mathbf{t}, \mathbf{s}) := \left(\frac{\pi^{1+(d_1+d_2)/2}}{\Gamma((1+d_1)/2)\Gamma((1+d_2)/2)} \|\mathbf{t}\|^{1+d_1} \|\mathbf{s}\|^{1+d_2}\right)^{-1},$$

$$f^n_{\mathbf{X},\mathbf{Y}}(\mathbf{t},\mathbf{s}) := \frac{1}{n}\sum_{j=1}^n \exp\left(i\mathbf{t}^\top \widehat{\mathbf{R}}^{\mathbf{X}}_n(\mathbf{X}_j) + i\mathbf{s}^\top \widehat{\mathbf{R}}^{\mathbf{Y}}_n(\mathbf{Y}_j)\right),$$

$$f^n_{\mathbf{X}}(\mathbf{t}) := \frac{1}{n}\sum_{j=1}^n \exp\left(i\mathbf{t}^\top \mathbf{h}^{d_1}_j\right) \qquad \text{and} \qquad f^n_{\mathbf{Y}}(\mathbf{s}) := \frac{1}{n}\sum_{j=1}^n \exp\left(i\mathbf{s}^\top \mathbf{h}^{d_2}_j\right)$$

for $\mathbf{t} \in \mathbb{R}^{d_1}$, $\mathbf{s} \in \mathbb{R}^{d_2}$ and $i = \sqrt{-1}$. Note that $f^n_{\mathbf{X}}(\cdot)$ and $f^n_{\mathbf{Y}}(\cdot)$ are deterministic quantities. Recall that, under H_0, $(\widehat{\mathbf{R}}^{\mathbf{X}}_n(\mathbf{X}_1), \ldots, \widehat{\mathbf{R}}^{\mathbf{X}}_n(\mathbf{X}_n))$ and $(\widehat{\mathbf{R}}^{\mathbf{Y}}_n(\mathbf{Y}_1), \ldots, \widehat{\mathbf{R}}^{\mathbf{Y}}_n(\mathbf{Y}_n))$ are independent and distributed uniformly over all $n!$ permutations of the sets $\mathcal{H}^{d_1}_n$ and $\mathcal{H}^{d_2}_n$ respectively (see Lemma 4.1 in the main paper). Let σ_1 and σ_2 be two independent random permutations of the set $\{1, 2, \ldots, n\}$. Then note that:

$$\begin{aligned} f^n_{\mathbf{X},\mathbf{Y}}(\mathbf{t},\mathbf{s}) &\overset{d}{=} \frac{1}{n}\sum_{j=1}^n \exp\left(i\mathbf{t}^\top \mathbf{h}^{d_1}_{\sigma_1(j)} + i\mathbf{s}^\top \mathbf{h}^{d_2}_{\sigma_2(j)}\right) \\ &\overset{d}{=} \frac{1}{n}\sum_{j=1}^n \exp\left(i\mathbf{t}^\top \mathbf{h}^{d_1}_{j} + i\mathbf{s}^\top \mathbf{h}^{d_2}_{\sigma_2(j)}\right). \end{aligned} \tag{F.24}$$

Using (F.24), along with [50, Theorem 1], we get:

$$\mathrm{RdCov}^2_n \overset{d}{=} \int \left| \frac{1}{n}\sum_{j=1}^n \exp\left(i\mathbf{t}^\top \mathbf{h}^{d_1}_{j} + i\mathbf{s}^\top \mathbf{h}^{d_2}_{\sigma_2(j)}\right) - f^n_{\mathbf{X}}(\mathbf{t}) f^n_{\mathbf{Y}}(\mathbf{s}) \right|^2 w(\mathbf{t},\mathbf{s})\, d\mathbf{t}\, d\mathbf{s}. \tag{F.25}$$

Lemma F.1 (see Appendix G.1 for a proof) below deals with the asymptotic behavior of permutation statistics of the same form as in the right hand side of (F.25).

Lemma F.1. *Consider two infinite deterministic sequences, $\{\mathbf{U}_i\}_{i\geq 1}$ and $\{\mathbf{V}_i\}_{i\geq 1}$, $\mathbf{U}_i \in [0,1]^{d_1}$ and $\mathbf{V}_i \in [0,1]^{d_2}$, such that $\mathbb{P}_n = (1/n)\sum_{i=1}^n \delta_{\mathbf{U}_i} \xrightarrow{w} \mathcal{U}^{d_1}$ and $\mathbb{Q}_n = (1/n)\sum_{i=1}^n \delta_{\mathbf{V}_i} \xrightarrow{w} \mathcal{U}^{d_2}$, where $\mathcal{U}^{d_1}$ (and $\mathcal{U}^{d_2}$) are the standard Lebesgue measures on $[0,1]^{d_1}$ (and $[0,1]^{d_2}$). Further, let S_n denote the set of all permutations of $\{1, 2, \ldots, n\}$ and π_n be a random permutation drawn uniformly from S_n. Define the following:*

$$\begin{aligned} \xi_n(\mathbf{t},\mathbf{s}) := \sqrt{n}\Bigg(&\frac{1}{n}\sum_{k=1}^n \exp\left(i\mathbf{t}^\top \mathbf{U}_k + i\mathbf{s}^\top \mathbf{V}_{\pi_n(k)}\right) - \left(\frac{1}{n}\sum_{k=1}^n \exp\left(i\mathbf{t}^\top \mathbf{U}_k\right)\right) \\ &\times \left(\frac{1}{n}\sum_{k=1}^n \exp\left(i\mathbf{s}^\top \mathbf{V}_k\right)\right)\Bigg) \end{aligned}$$

for $\mathbf{t} \in \mathbb{R}^{d_1}$ *and* $\mathbf{s} \in \mathbb{R}^{d_2}$*. Then, we have,*

$$D_n := \int_{\mathbb{R}^{d_1}\times\mathbb{R}^{d_2}} \left|\xi_n(\mathbf{t},\mathbf{s})\right|^2 w(\mathbf{t},\mathbf{s})\, d\mathbf{t}\, d\mathbf{s} \xrightarrow{w} \int_{\mathbb{R}^{d_1}\times\mathbb{R}^{d_2}} \left|\xi(\mathbf{t},\mathbf{s})\right|^2 w(\mathbf{t},\mathbf{s})\, d\mathbf{t}\, d\mathbf{s} := D. \tag{F.26}$$

Here $\xi(\cdot,\cdot)$ *is a complex-valued Gaussian process with mean* 0 *and covariance kernel*

$$R((\mathbf{t}_1,\mathbf{s}_1),(\mathbf{t}_0,\mathbf{s}_0)) := \left(f_{d_1}(\mathbf{t}_1-\mathbf{t}_0) - f_{d_1}(\mathbf{t}_1)\overline{f_{d_1}(\mathbf{t}_0)}\right)\left(f_{d_2}(\mathbf{s}_1-\mathbf{s}_0) - f_{d_2}(\mathbf{s}_1)\overline{f_{d_2}(\mathbf{s}_0)}\right)$$

where $f_{d_1}(\cdot)$ *and* $f_{d_2}(\cdot)$ *denote the characteristic functions of a* $\mathcal{U}^{d_1}$ *and a* $\mathcal{U}^{d_2}$ *random variable respectively.*

With the above lemma in mind, note that the empirical distributions on $\mathcal{H}_n^{d_1}$ and $\mathcal{H}_n^{d_2}$ converge weakly to $\mathcal{U}^{d_1}$ and $\mathcal{U}^{d_2}$ respectively (by assumption **(AP2)**). By setting $\{\mathbf{U}_j\}_{j\geq 1}$ as $\{\mathbf{h}_j^{d_1}\}_{j\geq 1}$ and $\{\mathbf{V}_j\}_{j\geq 1}$ as $\{\mathbf{h}_j^{d_2}\}_{j\geq 1}$, and applying Lemma F.1 to the right side of (F.25), we get that, as $n\to\infty$:

$$n\mathrm{RdCov}_n^2 \xrightarrow{w} D \stackrel{d}{=} \sum_{j\geq 1} \lambda_j Z_j^2 \tag{F.27}$$

where D is defined in (F.26), λ_j's are fixed positive constants, and Z_j's are independent standard normals. The last equivalence in (F.27) follows from [32, Chapter 1, Section 2] using standard Karhunen-Loève type expansions for Gaussian processes. This completes the proof. □

F.9 Proof of Theorem 4.2

Proof. Recall that $\mathrm{RdCov}_n^2 = S_1 + S_2 - 2S_3$ where

$$S_1 := \frac{1}{n^2}\sum_{k,l=1}^{n} \|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k) - \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l)\|\|\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_k) - \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_l)\|,$$

$$S_2 := \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k) - \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l)\|\right) \times \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_k) - \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_l)\|\right),$$

$$S_3 := \frac{1}{n^3}\sum_{k,l,m=1}^{n} \|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k) - \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l)\|\|\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_k) - \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_m)\|.$$

Let us focus on S_1. Observe that by the triangle inequality, for any $k, l \in \{1, 2, \ldots, n\}$:

$$\|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k) - \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l)\| \geq \|\mathbf{R}^{\mathbf{X}}(\mathbf{X}_k) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_l)\| - \|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_k)\| - \|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_l)\|. \tag{F.28}$$

Plugging in (F.28) in S_1, we get:

$$\begin{aligned}\liminf_{n\to\infty} S_1 \geq \liminf_{n\to\infty} &\left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\mathbf{R}^{\mathbf{X}}(\mathbf{X}_k) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_l)\| \|\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_k) - \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_l)\|\right) \\ &- \limsup_{n\to\infty} \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\mathbf{R}^{\mathbf{X}}(\mathbf{X}_k) - \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k)\| \|\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_k) - \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_l)\|\right) \\ &- \limsup_{n\to\infty} \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\mathbf{R}^{\mathbf{X}}(\mathbf{X}_l) - \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l)\| \|\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_k) - \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_l)\|\right). \end{aligned} \tag{F.29}$$

Now, by Theorem 2.1 the last two terms on the right side of (F.29) equal 0 a.s. Therefore,

$$\liminf_{n\to\infty} S_1 \geq \liminf_{n\to\infty} \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\mathbf{R}^{\mathbf{X}}(\mathbf{X}_k) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_l)\| \|\widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_k) - \widehat{\mathbf{R}}_n^{\mathbf{Y}}(\mathbf{Y}_l)\|\right) \quad \text{a.s.} \tag{F.30}$$

Next, starting from the right side of (F.30) and repeating the same argument as above on the $\mathbf{Y}$'s instead of $\mathbf{X}$'s, we get:

$$\liminf_{n\to\infty} S_1 \geq \liminf_{n\to\infty} \left(\frac{1}{n^2}\sum_{k,l=1}^{n} \|\mathbf{R}^{\mathbf{X}}(\mathbf{X}_k) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_l)\| \|\mathbf{R}^{\mathbf{Y}}(\mathbf{Y}_k) - \mathbf{R}^{\mathbf{Y}}(\mathbf{Y}_l)\|\right) \quad \text{a.s.} \tag{F.31}$$

Note that $\{\mathbf{R}^{\mathbf{X}}(\mathbf{X}_i), \mathbf{R}^{\mathbf{Y}}(\mathbf{Y}_i)\}_{1\leq i\leq n}$ are i.i.d. random vectors, which implies that the right side of (F.31) is a standard V-statistic. Consequently, by invoking the strong law of large numbers for V-statistics, we have:

$$\liminf_{n\to\infty} S_1 \geq \mathbb{E}\big[\|\mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^1) - \mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^2)\| \|\mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^1) - \mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^2)\|\big] \quad \text{a.s.} \tag{F.32}$$

where $(\mathbf{Z}_1^1, \mathbf{Z}_2^1), (\mathbf{Z}_1^2, \mathbf{Z}_2^2)$ are independent observations having the same distribution as $(\mathbf{X}_1, \mathbf{Y}_1)$. Also, analogous to (F.28), an application of the triangle inequality also yields the following:

$$\begin{aligned}&\|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k) - \widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l)\| \\ &\leq \|\mathbf{R}^{\mathbf{X}}(\mathbf{X}_k) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_l)\| + \|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_k) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_k)\| + \|\widehat{\mathbf{R}}_n^{\mathbf{X}}(\mathbf{X}_l) - \mathbf{R}^{\mathbf{X}}(\mathbf{X}_l)\|. \end{aligned} \tag{F.33}$$

Starting from (F.33), and repeating the same arguments as in (F.29), (F.30), (F.31) and (F.32), we get:

$$S_1 \overset{n\to\infty}{\longrightarrow} \mathbb{E}\big[\|\mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^1)\mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^2)\| \|\mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^1) - \mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^2)\|\big] \quad \text{a.s.} \tag{F.34}$$

Similar arguments applied to S_2 and S_3 yield the following:

$$S_2 \overset{n\to\infty}{\longrightarrow} \mathbb{E}\big[\|\mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^1) - \mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^2)\|\big]\mathbb{E}\big[\|\mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^1) - \mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^2)\|\big] \qquad \text{a.s.} \tag{F.35}$$

and,

$$S_3 \overset{n\to\infty}{\longrightarrow} \mathbb{E}\big[\|\mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^1) - \mathbf{R}^{\mathbf{X}}(\mathbf{Z}_1^2)\|\|\mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^1) - \mathbf{R}^{\mathbf{Y}}(\mathbf{Z}_2^3)\|\big] \qquad \text{a.s.} \tag{F.36}$$

where $(\mathbf{Z}_1^1, \mathbf{Z}_2^1), (\mathbf{Z}_1^2, \mathbf{Z}_2^2), (\mathbf{Z}_1^3, \mathbf{Z}_2^3)$ are independent observations having the same distribution as $(\mathbf{X}_1, \mathbf{Y}_1)$. Finally, by combining (F.34), (F.35), (F.36) and (3.4) in the main paper, we get:

$$\mathrm{RdCov}_n^2 = S_1 + S_2 - 2S_3 \overset{n\to\infty}{\longrightarrow} \mathrm{RdCov}^2(\mathbf{X}, \mathbf{Y}) \qquad \text{a.s.} \tag{F.37}$$

which completes the proof of the first part.

For the second part, note that, Theorem 4.1 in the main paper implies $c_n = O(1)$. Also, Lemma 3.1 (part (ii)) in the main paper and (F.37) imply that whenever $\mu \neq \mu_{\mathbf{X}} \otimes \mu_{\mathbf{Y}}$, we will have:

$$n\mathrm{RdCov}_n^2 \overset{n\to\infty}{\longrightarrow} \infty \qquad \text{a.s.}$$

which in turn, yields $\mathbb{P}(n\mathrm{RdCov}_n^2 > c_n) \overset{n\to\infty}{\longrightarrow} 1$ if $\mu \neq \mu_{\mathbf{X}} \otimes \mu_{\mathbf{Y}}$, thereby completing the proof. □

F.10 Proof of Lemma 4.3

Proof. $\mathrm{RE}_{m,n}^2$, as defined in (4.4) in the main paper, is a function of $(\widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{X}_1), \ldots, \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{X}_m), \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{Y}_1), \ldots, \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{Y}_n))$. Note that the above vector is uniformly distributed over the set of all $(m+n)!$ permutations of the fixed grid $\mathcal{H}_{m+n}^d$ under H_0 and **(AP3)** (see Proposition 2.2, part (ii) in the main paper). This results in the distribution-free property of the statistic $\mathrm{RE}_{m,n}^2$ (under H_0). □

F.11 Proof of Lemma 4.4

Proof. Let us first prove (4.6) in the main paper. Note that $\mathbb{P}\big(H_\lambda^{X,Y}(X) \leq t\big) = F^X\big((H_\lambda^{X,Y})^{-1}(t)\big)$ and $\mathbb{P}\big(H_\lambda^{X,Y}(Y) \leq t\big) = G^Y\big((H_\lambda^{X,Y})^{-1}(t)\big)$ for all t in the support of $H_\lambda^{X,Y}$. By (3.5) in the main paper, we have:

$$\begin{aligned}\frac{1}{2}\mathrm{RE}_\lambda^2(X,Y) &= \int_0^1 \Big(\mathbb{P}(H_\lambda^{X,Y}(X) \leq t) - \mathbb{P}(H_\lambda^{X,Y}(Y) \leq t)\Big)^2 dt \\ &= \int_0^1 \Big(F^X\big((H_\lambda^{X,Y})^{-1}(t)\big) - G^Y\big((H_\lambda^{X,Y})^{-1}(t)\big)\Big)^2 dt\end{aligned}$$

$$= \int_{-\infty}^{\infty} \left(F^X(t) - G^Y(t) \right)^2 dH_\lambda^{X,Y}(t)$$

where the last line follows by a simple change of variable argument. This proves (4.6) in the main paper.

In order to prove (4.5) in the main paper, note that, by using [4, Equation 5], we get:

$$\frac{1}{2}\mathrm{RE}_{m,n}^2 = \int_{-\infty}^{\infty} \left(\frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\big(\widehat{R}_{m,n}^{X,Y}(X_j) \le t\big) - \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\big(\widehat{R}_{m,n}^{X,Y}(Y_j) \le t\big) \right)^2 dt. \tag{F.38}$$

Clearly, for $t < (m+n)^{-1}$ or $t > 1$, the right hand side of (F.38) equals 0. For $t \in [k(m+n)^{-1}, (k+1)(m+n)^{-1})$, $k \in \{1, 2, \ldots, m+n-1\}$, observe that:

$$\frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\big(\widehat{R}_{m,n}^{X,Y}(X_j) \le t\big) = \frac{1}{m} \sum_{j=1}^{m} \mathbf{1}(X_j \le Z_{(k)}) \tag{F.39}$$

where $Z_{(k)}$ denotes the k'th order statistic of the pooled sample $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$. A similar observation as in (F.39) also holds for the Y_j's. Consequently, by adding up over $k \in \{1, 2, \ldots, m+n\}$, we get that the right hand side of (F.38) equals

$$\begin{aligned} &\frac{1}{m+n} \sum_{i=1}^{m+n} \left(\frac{1}{m} \sum_{j=1}^{m} \mathbf{1}(X_j \le Z_{(i)}) - \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}(Y_j \le Z_{(i)}) \right)^2 \\ &= \int_{-\infty}^{\infty} (F_m^X(t) - G_n^Y(t))^2 \, dH_{m+n}(t) \end{aligned}$$

which completes the proof of (4.5) in the main paper. □

F.12 Proof of Theorem 4.3

Proof. First we start with some notation: set $\mathcal{H}_{m+n}^d := \{\mathbf{h}_1^d, \mathbf{h}_2^d, \ldots, \mathbf{h}_{m+n}^d\}$ and

$$F_m^{\mathbf{a}}(r) := \frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\left(\mathbf{a}^\top \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{X}_j) \le r \right) \quad \text{and} \quad G_n^{\mathbf{a}}(r) := \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\left(\mathbf{a}^\top \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{Y}_j) \le r \right) \tag{F.40}$$

where $\mathbf{a} \in \mathcal{S}^{d-1} = \{\mathbf{z} : \|\mathbf{z}\| = 1\}$, $r \in \mathbb{R}$ and $\mathbf{1}(\cdot)$ denotes the standard indicator function. Recall that $\kappa(\cdot)$ is the uniform measure on $\mathcal{S}^{d-1}$ and $\gamma_d = (2\Gamma(d/2))^{-1}\sqrt{\pi}(d-1)\Gamma((d-1)/2)$ for $d > 1$ and $\gamma_d = 1$ for $d = 1$.

Next, recall that $(\widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{X}_1), \ldots, \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{X}_m), \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{Y}_1), \ldots, \widehat{\mathbf{R}}_{m,n}^{\mathbf{X},\mathbf{Y}}(\mathbf{Y}_n))$ is uniformly distributed over the set of all $(m+n)!$ permutations of the set $\mathcal{H}_{m+n}^d$ (see the proof of Lemma 4.3

in Appendix F.10). From (F.40) in the main paper, this implies the following:

$$(F_m^{\mathbf{a}}(r), G_n^{\mathbf{a}}(r)) \stackrel{d}{=} \left(\frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\big(\mathbf{a}^\top \mathbf{h}^d_{\sigma_1(j)} \le r\big), \frac{1}{n} \sum_{j=m+1}^{m+n} \mathbf{1}\big(\mathbf{a}^\top \mathbf{h}^d_{\sigma_1(j)} \le r\big) \right)$$

for a random permutation σ_1 of the set $\{1, 2, \ldots, m+n\}$. Now, by [4, Lemma 2.3, Equation 5], we further get:

$$\begin{aligned} &\frac{mn}{m+n} \mathrm{RE}^2_{m,n} \\ &\stackrel{d}{=} \frac{mn}{m+n} \cdot \gamma_d \int_{\mathcal{S}^{d-1}} \int_{-\infty}^{\infty} \left(\frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\big(\mathbf{a}^\top \mathbf{h}^d_{\sigma_1(j)} \le r\big) - \frac{1}{n} \sum_{j=m+1}^{m+n} \mathbf{1}\big(\mathbf{a}^\top \mathbf{h}^d_{\sigma_1(j)} \le r\big) \right)^2 dr \, d\kappa(\mathbf{a}). \end{aligned} \tag{F.41}$$

Let us take a look at Lemma F.2 below (see Appendix G.2 for a proof) which provides a general result on permutation statistics of the same form as in the right side of (F.41).

Lemma F.2. *Consider an infinite sequence $\{\mathbf{U}_i\}_{i \ge 1}$, $\mathbf{U}_i \in \mathbb{R}^d$, such that $\mathbb{P}_{m+n} = (m+n)^{-1} \sum_{i=1}^{m+n} \delta_{\mathbf{U}_i} \xrightarrow{w} \mathcal{U}^d$ (uniform distribution on $[0,1]^d$), as $\min(m,n) \to \infty$. Further, let S_N denote the set of all permutations of $\{1, 2, \ldots, N\}$ and π_N be a random permutation drawn uniformly from S_N. Define $\mathcal{S}^{d-1} = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| = 1\}$ and $\Theta_{m,n} : \mathcal{S}^{d-1} \times \mathbb{R} \mapsto \mathbb{R}$ by,*

$$\Theta_{m,n}(\mathbf{a}, r) = \sqrt{\frac{mn}{m+n}} \cdot \left(\frac{1}{m} \sum_{i=1}^{m} \mathbf{1}(\mathbf{a}^\top \mathbf{U}_{\pi_N(i)} \le r) - \frac{1}{n} \sum_{j=m+1}^{m+n} \mathbf{1}(\mathbf{a}^\top \mathbf{U}_{\pi_N(j)} \le r) \right).$$

Then we have,

$$E_{m,n} := \int_{\mathcal{S}^{d-1}} \int_{\mathbb{R}} \Theta^2_{m,n}(\mathbf{a}, r) \, dr \, d\kappa(\mathbf{a}) \xrightarrow{w} \int_{\mathcal{S}^{d-1}} \int_{\mathbb{R}} \Theta^2(\mathbf{a}, r) \, dr \, d\kappa(\mathbf{a}) := E$$

as $\min(m,n) \to \infty$. Here $\kappa(\cdot)$ denotes the uniform measure on $\mathcal{S}^{d-1}$ and $\Theta(\cdot,\cdot)$ is a mean 0 Gaussian process with covariance kernel given by,

$$C\big((\mathbf{a}_1, r_1), (\mathbf{a}_2, r_2)\big) := \mathbb{P}(\mathbf{a}_1^\top \widetilde{\mathbf{U}} \le r_1, \mathbf{a}_2^\top \widetilde{\mathbf{U}} \le r_2) - \mathbb{P}(\mathbf{a}_1^\top \widetilde{\mathbf{U}} \le r_1) \cdot \mathbb{P}(\mathbf{a}_2^\top \widetilde{\mathbf{U}} \le r_2), \tag{F.42}$$

for $(\mathbf{a}_1, r_1)$, $(\mathbf{a}_2, r_2) \in \mathcal{S}^{d-1} \times \mathbb{R}$, where $\widetilde{\mathbf{U}}$ has the same distribution as $\mathcal{U}^d$.

With the above lemma in mind, note that the empirical distribution on $\mathcal{H}_n^d$ converges weakly to $\mathcal{U}^d$ under assumption **(AP4)**. By setting $\{\mathbf{U}_j\}_{j\geq 1}$ as $\{\mathbf{h}_j^d\}_{j\geq 1}$, and applying Lemma F.2 to the right hand side of (F.41), we get that, as $\min(m,n)\to\infty$,

$$\frac{mn}{m+n}\mathrm{RE}_{m,n}^2 \xrightarrow{w} \gamma_d E \stackrel{d}{=} \sum_{j=1}^{\infty}\tau_j Z_j^2 \tag{F.43}$$

where τ_j's are fixed nonnegative constants and Z_j's are i.i.d. standard normals. The last equality in (F.43) follows from [32, Chapter 1, Section 2] using standard Karhunen-Loève type expansions of Gaussian processes. This completes the proof. □

F.13 Proof of Theorem 4.4

Proof. Note that $\mathrm{RE}_{m,n}^2$ is the average of products of Euclidean distances between bounded random vectors, in the same spirit as RdCov_n^2. Therefore, The proof is exactly similar to the proof of Theorem 4.2 in the main paper in Appendix F.9. We leave the details to the reader. □

F.14 Proof of Proposition E.1

Proof. By the same argument as in Appendix F.1, we get that the rank vectors $(\widehat{\mathbf{R}}_n^j(\mathbf{X}_1^j),\ldots,\widehat{\mathbf{R}}_n^j(\mathbf{X}_n^j))$ are uniformly distributed over all $n!$ possible permutations of the set $\mathcal{H}_n^{d_j}$. Also these rank vectors are independent over $j\in\{1,2,\ldots,K\}$ under H_0. This implies that our test statistic, which is a function of $(\widehat{\mathbf{R}}_n^j(\mathbf{X}_1^j),\ldots,\widehat{\mathbf{R}}_n^j(\mathbf{X}_n^j))_{1\leq j\leq K}$, is distribution-free under H_0. Consequently, the threshold c_n, is also distribution-free under H_0.

Finally, by Theorem 4.2 in the main paper, we see that $\mathrm{RdCov}_{n,j}^2$ converges a.s. to the following quantity:

$$\mathrm{RdCov}_n^2 \xrightarrow{a.s.} \sum_{j=1}^{K-1}\mathrm{RdCov}_*^2(\mathbf{X}^j,\mathbf{X}^{j+}) \tag{F.44}$$

where $\mathrm{RdCov}_*^2(\mathbf{X}^j,\mathbf{X}^{j+})$ is the usual distance covariance between $\mathbf{R}_j(\mathbf{X}^j)$ and $(\mathbf{R}_{j+1}(\mathbf{X}^{j+1}),\ldots,\mathbf{R}_k(\mathbf{X}^k))$ and $R_j(\cdot)$ is the population rank map for μ_j, $1\leq j\leq K$.

Next, note that under H_0, $n\mathrm{RdCov}_{n,j}^2=\mathcal{O}_p(1)$ for $1\leq j\leq K-1$ by Theorem 4.1 in the main paper (the same second moment computations as in (G.1) and (G.3)). Therefore, $n\mathrm{RdCov}_n^2=\mathcal{O}_p(1)$ which implies $c_n=\mathcal{O}(1)$.

Now, under H_1, we will show that the right side of (F.44) is strictly positive. In this direction, let us proceed by contradiction. Suppose that the right side of (F.44) equals 0 under H_1. This implies $\mathrm{RdCov}_*^2(\mathbf{X}^j,\mathbf{X}^{j+})=0$ for all $1\leq j\leq K-1$. Therefore, by the same argument as in Lemma 3.1 (part (b)) in the main paper, $\mathbf{X}^j$ and $\mathbf{X}^{j+}$ are independent for $1\leq j\leq K-1$.

An alternate way of stating this would be as follows:

$$\mathbb{E}\left[\exp\left(i\sum_{l=j}^{K}\mathbf{t}_l^\top\mathbf{X}^l\right)\right] = \mathbb{E}\left[\exp\left(i\mathbf{t}_j^\top\mathbf{X}^j\right)\right] \times \mathbb{E}\left[\exp\left(i\sum_{l=j+1}^{K}\mathbf{t}_l^\top\mathbf{X}^l\right)\right] \tag{F.45}$$

for all $(\mathbf{t}^1,\ldots,\mathbf{t}_K) \in \mathbb{R}^{d_1}\times\ldots\times\mathbb{R}^{d_k}$ and $1 \le j \le K-1$. Next, by [33], we have:

$$\begin{aligned}&\left|\mathbb{E}\left[\exp\left(i\sum_{l=1}^{K}\mathbf{t}_l^\top\mathbf{X}^l\right)\right] - \prod_{l=1}^{K}\mathbb{E}\left[\exp\left(i\mathbf{t}_l^\top\mathbf{X}^l\right)\right]\right| \\ &\le \sum_{j=1}^{K-1}\left|\mathbb{E}\left[\exp\left(i\sum_{l=j}^{K}\mathbf{t}_l^\top\mathbf{X}^l\right)\right] - \mathbb{E}\left[\exp\left(i\mathbf{t}_j^\top\mathbf{X}^j\right)\right] \times \mathbb{E}\left[\exp\left(i\sum_{l=j+1}^{K}\mathbf{t}_l^\top\mathbf{X}^l\right)\right]\right|. \end{aligned} \tag{F.46}$$

The right hand side of (F.46) equals 0 by (F.45), and consequently the left hand side of (F.46) equals 0. Therefore $\mu = \mu_1\otimes\ldots\otimes\mu_K$, which contradicts H_1 and consequently proves the claim. As a result we also have $n\mathrm{RdCov}_n^2 \xrightarrow{a.s.} \infty$ under H_1. As $c_n = \mathcal{O}(1)$, we have

$$\mathbb{P}(n\mathrm{RdCov}_n^2 \ge c_n) \xrightarrow{n\to\infty} 1$$

which completes the proof of consistency.

F.15 Proof of Proposition E.2

Note that the set of n vectors $(\widehat{\mathbf{R}}_n(\mathbf{X}_1^j),\ldots,\widehat{\mathbf{R}}_n(\mathbf{X}_{n_j}^j))_{1\le j\le K}$ are uniformly distributed over the $n!$ permutations of the set $\mathcal{H}_n^d$ (same argument as in Proposition 2.2). This implies that the test statistic $\mathrm{RE}_{1:K,n}^2$ is distribution-free and consequently, so is c_n.

By the same argument as in the proof of Theorem 4.4, we have:

$$\mathrm{RE}_{1:K,n}^2 = \sum_{j=1}^{K-1}\mathrm{RE}_{j:(j+1),n}^2 \xrightarrow{a.s.} \sum_{j=1}^{K-1}\mathrm{RE}_{\lambda,*}^2(\mathbf{X}^j,\mathbf{X}^{j+1}) \tag{F.47}$$

where $\mathrm{RE}_{\lambda,*}^2(\mathbf{X}^j,\mathbf{X}^{j+1})$ is the usual energy distance between $\mathbf{R}_\lambda(\mathbf{X}^j)$ and $\mathbf{R}_\lambda(\mathbf{X}^{j+1})$, $\mathbf{R}_\lambda(\cdot)$ denotes the population rank map corresponding to the measure $\sum_{j=1}^K\lambda_j\mu_j$.

Under H_0, by similar second moment calculations as in the proof of Theorem 4.3, we have $n\mathrm{RE}_{1:K,n}^2 = \mathcal{O}_p(1)$ which in turn implies $c_n = \mathcal{O}(1)$.

Under H_1, there exists $\widetilde{j} \le K-1$ such that $\mu_{\widetilde{j}} \ne \mu_{\widetilde{j}+1}$ which implies $\mathrm{RE}_{\lambda,*}^2(\mathbf{X}^{\widetilde{j}},\mathbf{X}^{\widetilde{j}+1}) > 0$ and consequently the right hand side of (F.47) is strictly positive. Therefore, $n\mathrm{RE}_{1:K,n}^2 \xrightarrow{a.s.} \infty$ and as $c_n = \mathcal{O}(1)$, we have

$$\mathbb{P}(n\mathrm{RE}_{1:K,n}^2 \ge c_n) \xrightarrow{n\to\infty} 1$$

which completes the proof of consistency. □

G Some general results on permutation statistics

In this section, we will prove some general results on asymptotic distributions of certain permutation based statistics which were used in Appendix F. Since the distinction between vectors and scalars is pretty clear in this section, we will drop the boldface fonts (previously used to denote vectors) subsequently for notational convenience.

G.1 Proof of Lemma F.1

Recall the notation in Lemma F.1. Note that D (in (F.26)) is well-defined (see [32, Chapter 1, Section 2]) and has finite expectation. For any $c > 0$, define the following:

$$W_{n,c} := \int_{1/c \le \|t\|, \|s\| \le c} \left|\xi_n(t,s)\right|^2 w(t,s)\, dt\, ds, \quad W_c = \int_{1/c \le \|t\|, \|s\| \le c} \left|\xi(t,s)\right|^2 w(t,s)\, dt\, ds.$$

Our proof will proceed through the following three steps:

Lemma G.1. *For any $\delta > 0$, $\lim_{c\to\infty} \limsup_{n\to\infty} \mathbb{P}(|W_{n,c} - D_n| > \delta) = 0$.*

Proposition G.1. *For any $\delta > 0$, $\lim_{c\to\infty} \mathbb{P}(|W_c - D| > \delta) = 0$.*

Lemma G.2. *For any $c > 0$, $W_{n,c} \xrightarrow{w} W_c$ as $n \to \infty$.*

Combining Proposition G.1, Lemmas G.1 and G.2 with [47, Lemma 2.5], yields $D_n \xrightarrow{w} D$ as $n \to \infty$ and completes the proof.

Proof of Lemma G.2. For $(t,s) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$, define:

$$f_U^n(t) := \frac{1}{n}\sum_{k=1}^{n} \exp\left(it^\top U_k\right) \quad , \quad f_V^n(t) := \frac{1}{n}\sum_{k=1}^{n} \exp\left(it^\top V_k\right)$$

In the following, U_k's and V_k's are fixed, so the expectations are taken with respect to the randomness arising out of the randomly drawn permutation $\pi_n \in S_n$. Note that, for each

$(t_1, s_1), (t_0, s_0)$, we have $\mathbb{E}[\xi_n(t_1, s_1)] = 0$ and

$$\begin{aligned}
\mathbb{E}\left[\xi_n(t_1, s_1)\overline{\xi_n(t_0, s_0)}\right] &= \frac{n}{n-1}\left(f_U^n(t_1 - t_0) - f_U^n(t_1)\overline{f_U^n(t_0)}\right)\left(f_V^n(s_1 - s_0) - f_V^n(s_1)\overline{f_V^n(s_0)}\right) \\
&\overset{n\to\infty}{\longrightarrow} \left(f_{d_1}(t_1 - t_0) - f_{d_1}(t_1)\overline{f_{d_1}(t_0)}\right)\left(f_{d_2}(s_1 - s_0) - f_{d_2}(s_1)\overline{f_{d_2}(s_0)}\right) \\
&= R((t_1, s_1), (t_0, s_0)) \qquad \text{(G.1)}
\end{aligned}$$

where the convergence follows from the assumption that $\mathbb{P}_n \xrightarrow{w} \mathcal{U}^{d_1}$ and $\mathbb{Q}_n \xrightarrow{w} \mathcal{U}^{d_2}$.

Now fix $M \geq 1$, and two sequences of real numbers $(\alpha_1, \ldots, \alpha_M)$ and $(\beta_1, \ldots, \beta_M)$. For $\{(t_m, s_m) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}\}_{m=1}^M$, define:

$$\begin{aligned}
\Lambda_n(t_m, s_m) &\coloneqq \sqrt{n}\left[\frac{1}{n}\sum_{k=1}^n \cos\left(t_m^\top U_k + s_m^\top V_{\pi_n(k)}\right) - \frac{1}{n^2}\sum_{k,l}\cos\left(t_m^\top U_k + s_m^\top V_l\right)\right] \\
\Theta_n(t_m, s_m) &\coloneqq \sqrt{n}\left[\frac{1}{n}\sum_{k=1}^n \sin\left(t_m^\top U_k + s_m^\top V_{\pi_n(k)}\right) - \frac{1}{n^2}\sum_{k,l}\sin\left(t_m^\top U_k + s_m^\top V_l\right)\right].
\end{aligned}$$

Note that Λ_n and Θ_n are centered random variables. Further, define two matrices $(A_{kl})_{n\times n}$ and $(C_{kl})_{n\times n}$ as:

$$\begin{aligned}
A_{kl} &\coloneqq \frac{1}{\sqrt{n}}\sum_{m=1}^M \left[\alpha_m \cos\left(t_m^\top U_k + s_m^\top V_l\right) + \alpha_m \sin\left(t_m^\top U_k + s_m^\top V_l\right)\right], \\
C_{kl} &\coloneqq A_{kl} - \overline{A_{k\cdot}} - \overline{A_{\cdot l}} + A_{\cdot\cdot},
\end{aligned}$$

and note that $\sum_{k=1}^n C_{k\pi_n(k)} = \sum_{m=1}^M \left(\alpha_m \Lambda_n(t_m, s_m) + \beta_m \Theta_n(t_m, s_m)\right)$. By Hoeffding's Combinatorial Central Limit theorem (see e.g., [11, Theorem 1.1]), we will have $\sum_{k=1}^n C_{k\pi_n(k)} \xrightarrow{w} \mathcal{N}(0, \sigma^2)$ if the following two conditions hold: (a) $\sum_{k,l}|C_{kl}|^3 = o(n)$ and (b) $\mathrm{Var}\left(\sum_{k=1}^n C_{k\pi_n(k)}\right) \overset{n\to\infty}{\longrightarrow} \sigma^2$. As sine and cosine functions are bounded, and U_k's, V_k's all lie in fixed (free of n) compact sets, it is easy to check that $\sum_{k,l}|C_{ij}|^3 = O\left(\sqrt{n}\right)$, so condition (a) is satisfied. For verifying (b), let us first set up some prerequisites. Observe that:

$$\begin{aligned}
&\mathbb{E}\left[\Lambda_n(t_1, s_1)\Theta_n(t_0, s_0)\right] \\
&= \frac{1}{n}\sum_{k,l}\mathrm{Cov}\Big(\cos\left(t_1^\top U_k + s_1^\top V_{\pi_n(k)}\right), \sin\left(t_0^\top U_l + s_0^\top V_{\pi_n(l)}\right)\Big) \\
&= \frac{1}{n^2}\sum_{k,l}\cos\left(t_1^\top U_k + s_1^\top V_l\right)\sin\left(t_0^\top U_k + s_0^\top V_l\right) + \frac{1}{n^3(n-1)}\sum_{k\neq l, p\neq q}\cos\left(t_1^\top U_k + s_1^\top V_p\right) \\
&\qquad \times \sin\left(t_0^\top U_l + s_0^\top V_q\right) - \frac{1}{n^3}\sum_{p,k\neq l}\cos\left(t_1^\top U_k + s_1^\top V_p\right)\sin\left(t_0^\top U_l + s_0^\top V_p\right)
\end{aligned}$$

$$
\begin{aligned}
&- \frac{1}{n^3} \sum_{k,p\neq q} \cos\left(t_1^\top U_k + s_1^\top V_p\right) \sin\left(t_0^\top U_k + s_0^\top V_q\right) \\
&\overset{n\to\infty}{\longrightarrow} \mathbb{E}\Big[\cos\left(t_1^\top \widetilde{U}_1 + s_1^\top \widetilde{V}_1\right) \sin\left(t_0^\top \widetilde{U}_1 + s_0^\top \widetilde{V}_1\right) + \cos\left(t_1^\top \widetilde{U}_1 + s_1^\top \widetilde{V}_1\right) \sin\left(t_0^\top \widetilde{U}_2 + s_0^\top \widetilde{V}_2\right) \\
&\quad - \cos\left(t_1^\top \widetilde{U}_1 + s_1^\top \widetilde{V}_1\right) \sin\left(t_0^\top \widetilde{U}_1 + s_0^\top \widetilde{V}_2\right) - \cos\left(t_1^\top \widetilde{U}_1 + s_1^\top \widetilde{V}_1\right) \sin\left(t_0^\top \widetilde{U}_2 + s_1^\top \widetilde{V}_1\right)\Big].
\end{aligned}
$$

Here $\widetilde{U}_1, \widetilde{U}_2$ are i.i.d. $\mathcal{U}^{d_1}$ and $\widetilde{V}_1, \widetilde{V}_2$ are i.i.d. $\mathcal{U}^{d_2}$. A similar calculation shows that, for any $m_1, m_2 \in \{1, 2, \ldots, M\}$, $\mathbb{E}[\Lambda_n^2(t_{m_1}, s_{m_1})]$, $\mathbb{E}[\Theta_n^2(t_{m_1}, s_{m_1})]$, $\mathbb{E}[\Theta_n(t_{m_1}, s_{m_1})\Theta_n(t_{m_2}, s_{m_2})]$, $\mathbb{E}[\Lambda_n(t_{m_1}, s_{m_1})\Lambda_n(t_{m_2}, s_{m_2})]$ and $\mathbb{E}[\Lambda_n(t_{m_1}, s_{m_1})\Theta_n(t_{m_2}, s_{m_2})]$ also converges. Denote the corresponding limits as $\Lambda_{m_1}^2$, $\Theta_{m_1}^2$, $\Theta_{m_1 m_2}$, $\Lambda_{m_1 m_2}$ and $\Lambda_{m_1}\Theta_{m_2}$ respectively. This implies that $\mathrm{Var}\big(\sum_{k=1}^n C_{k\pi_n(k)}\big)$ equals,

$$
\begin{aligned}
&\mathbb{E}\Bigg[\sum_{m=1}^M \alpha_m^2 \Lambda_n^2(t_m, s_m) + \sum_{m=1}^M \beta_m^2 \Theta_n^2(t_m, s_m) + \sum_{m_1\neq m_2} \alpha_{m_1}\alpha_{m_2}\Lambda_n(t_{m_1}, s_{m_1})\Lambda_n(t_{m_2}, s_{m_2}) \\
&+ \sum_{m_1\neq m_2} \beta_{m_1}\beta_{m_2}\Theta_n(t_{m_1}, s_{m_1})\Theta_n(t_{m_2}, s_{m_2}) + \sum_{m_1, m_2} \alpha_{m_1}\beta_{m_2}\Lambda_n(t_{m_1}, s_{m_1})\Theta_n(t_{m_2}, s_{m_2}) \Bigg] \\
&\overset{n\to\infty}{\longrightarrow} \sum_{m=1}^M \alpha_m^2 \Lambda_m^2 + \sum_{m=1}^M \beta_m^2 \Theta_m^2 + \sum_{m_1\neq m_2} \alpha_{m_1}\alpha_{m_2}\Lambda_{m_1 m_2} + \sum_{m_1\neq m_2} \beta_{m_1}\beta_{m_2}\Theta_{m_1 m_2} \\
&\qquad + \sum_{m_1, m_2} \alpha_{m_1}\beta_{m_2}\Lambda_{m_1}\Theta_{m_2}.
\end{aligned}
$$

This completes the proof of (b) and therefore, $\sum_{k=1}^n C_{k\pi_n(k)}$ converges to a Gaussian limit. By the Cramér-Wold theorem, the vector

$$
\Gamma_n := (\Lambda_n(t_1, s_1), \ldots, \Lambda_n(t_M, s_M), \Theta_n(t_1, s_1), \ldots, \Theta_n(t_M, s_M))
$$

converges to a multivariate Gaussian distribution. Take $f : \mathbb{R}^{2M} \to \mathbb{C}^M$ such that $f(x_1, \ldots, x_{2M}) = (x_1 + ix_{M+1}, \ldots, x_M + ix_{2M})$ where $i = \sqrt{-1}$. Then, by the continuous mapping theorem, $f(\Gamma_n) = (\xi_n(t_1, s_1), \ldots, \xi_n(t_M, s_M))$ converges to a complex-valued Gaussian process with covariance kernel given by $R(\cdot, \cdot)$, as shown in (G.1).
For $c > 0$, define $A_c = \{t \in \mathbb{R}^{d_1} : (1/c) \leq \|t\| \leq c\}$ and $B_c = \{s \in \mathbb{R}^{d_2} : (1/c) \leq \|s\| \leq c\}$. Note that $A_c \times B_c$ is compact. The preceding discussion yields the convergence for the finite dimensional distributions of the process $\xi_n(\cdot, \cdot)$ over $A_c \times B_c$. In order to show $\xi_n(\cdot, \cdot) \overset{w}{\longrightarrow} \xi(\cdot, \cdot)$ in $L^\infty[A_c \times B_c]$ (in the sense of [27]), we would need to show asymptotic equicontinuity (see [27]), i.e., given any $\epsilon > 0$,

$$
\lim_{\delta\to 0} \limsup_{n\to\infty} \mathbb{P}^* \left(\sup_{\substack{s_1, s_0 \in B_c,\ t_1, t_0 \in A_c, \\ \sqrt{\|s_1 - s_0\|^2 + \|t_1 - t_0\|^2} \leq \delta}} \left| \xi_n(t_1, s_1) - \xi_n(t_0, s_0) \right| > \epsilon \right) = 0 \tag{G.2}
$$

where $\mathbb{P}^*$ denotes the outer probability. Note that $\xi_n(t,s) = \sum_{k=1}^n Z_{nk}(t,s)$ where $Z_{nk}(t,s) := n^{-1/2}\big[\exp\big(it^\top U_k + is^\top V_{\pi_n(k)}\big) - f_U^n(t) f_V^n(s)\big]$. Let $\rho((t_1,s_1),(t_0,s_0)) := \sqrt{\|t_1-t_0\|^2 + \|s_1-s_0\|^2}$. From the proof of [52, Theorem 2.11.1], (G.2) follows if we can show the following:

Step 1. There exists a sequence $\eta_n \downarrow 0$ such that $\sup_{1\le k\le n}\sup_{(t,s)\in A_c\times B_c}\big|Z_{nk}(t,s)\big| \le \eta_n$.

Step 2. For any sequence $\delta_n \downarrow 0$, we have:

$$\sup_{s_1,s_0\in B_c,\ t_1,t_0\in A_c, \rho((t_1,s_1),(t_0,s_0))\le\delta_n} \mathbb{E}^*\Big|\sum_{k=1}^n \big[Z_{nk}(t_1,s_1) - Z_{nk}(t_0,s_0)\big]\Big|^2 \overset{n\to\infty}{\longrightarrow} 0.$$

Step 3. For any sequence $\delta_n \downarrow 0$, $\int_0^{\delta_n}\sqrt{\log N(\epsilon, A_c\times B_c, d_n(\cdot,\cdot))}\,d\epsilon \overset{\mathbb{P}^*}{\longrightarrow} 0$ where $N(\epsilon, A_c\times B_c, d_n(\cdot,\cdot))$ denotes the ϵ covering number of the set $A_c\times B_c$ based on the random metric $d_n(\cdot,\cdot)$ satisfying $d_n^2((t_1,s_1),(t_0,s_0)) = \sum_{k=1}^n |Z_{nk}(t_1,s_1) - Z_{nk}(t_0,s_0)|^2$ (for related definitions in this context, see [52, Chapter 2.2]).

Note that $|Z_{nk}(\cdot,\cdot)| \le 2n^{-1/2}$ a.s. and so **step 1** holds. Note that, for any $s_1,s_0\in B_c$, $t_1,t_0\in A_c$ and $\rho((t_1,s_1),(t_0,s_0)) \le \delta_n$, we have:

$$\begin{aligned}
&\mathbb{E}^*\Big|\sum_{k=1}^n \big[Z_{nk}(t_1,s_1) - Z_{nk}(t_0,s_0)\big]\Big|^2 \\
&= \frac{n}{n-1}\Bigg[\Big(1 - f_U^n(t_0-t_1) f_V^n(s_0-s_1)\Big) + \Big(1 - f_U^n(t_1-t_0) f_V^n(s_1-s_0)\Big) \\
&\quad + \Big(f_U^n(t_0) f_V^n(s_0)\big(\overline{f_U^n(t_0) f_V^n(s_0)} - \overline{f_U^n(t_1) f_V^n(s_1)}\big)\Big) + \Big(f_U^n(t_1) f_V^n(s_1)\big(\overline{f_U^n(t_1) f_V^n(s_1)} \\
&\quad - \overline{f_U^n(t_0) f_V^n(s_0)}\big)\Big) - \Big(f_U^n(t_1)\big(\overline{f_U^n(t_1)} - \overline{f_V^n(s_1-s_0) f_U^n(t_0)}\big)\Big) \\
&\quad - \Big(f_U^n(t_0)\big(\overline{f_U^n(t_0)} - \overline{f_V^n(s_0-s_1) f_U^n(t_1)}\big)\Big) - \Big(f_V^n(s_1)\big(\overline{f_V^n(s_1)} - \overline{f_U^n(t_1-t_0) f_V^n(s_0)}\big)\Big) \\
&\quad - \Big(f_V^n(s_0)\big(\overline{f_V^n(s_0)} - \overline{f_U^n(t_0-t_1) f_V^n(s_1)}\big)\Big)\Bigg]. \qquad \text{(G.3)}
\end{aligned}$$

As $\sin(\cdot)$ and $\cos(\cdot)$ are Lipschitz functions with Lipschitz norm bounded by 1, each term within a parenthesis on the right hand side of (G.3) may be bounded in modulus by $4\delta_n$. This completes the proof of **step 2**. Further, once again using the Lipschitz nature of $\cos(\cdot)$ and $\sin(\cdot)$, it is easy to check that:

$$d_n((t_1,s_1),(t_0,s_0)) = \sqrt{\sum_{k=1}^n |Z_{nk}(t_1,s_1) - Z_{nk}(t_0,s_0)|^2} \le 10\rho((t_1,s_1),(t_0,s_0)) \qquad \text{(G.4)}$$

where the last event happens with $\mathbb{P}^*$ outer probability 1. By using $\lesssim_c$ to hide constants that only depend on p, q and c, we get the following chain of inequalities:

$$\int_0^{\delta_n} \sqrt{\log N(\epsilon, A_c \times B_c, d_n(\cdot,\cdot))}\, d\epsilon \overset{(a)}{\leq} \int_0^{\delta_n} \sqrt{\log N(\epsilon/10, A_c \times B_c, \rho(\cdot,\cdot))}\, d\epsilon$$
$$\overset{(b)}{\lesssim_c} \int_0^{\delta_n} \epsilon^{-1/2}\, d\epsilon \overset{n\to\infty}{\longrightarrow} 0$$

where (a) happens with $\mathbb{P}^*$ outer probability 1 and follows from (G.4), (b) follows from a standard volumetric argument for estimating covering numbers, see e.g., [51, Lemma 4.5]. This completes the proof of **step 3**. Therefore, by combining **steps 1, 2 and 3**, we get $\xi_n(\cdot,\cdot) \overset{w}{\longrightarrow} \xi(\cdot,\cdot)$ in $L^\infty[A_c \times B_c]$. Finally, note that $w(\cdot,\cdot)$ is bounded in $A_c \times B_c$. Lemma G.2 then follows by the continuous mapping theorem with the integral (over $A_c \times B_c$) operator. □

Proof of Lemma G.1. Define $\mathcal{B}_{d_1}(1) = \{z \in \mathbb{R}^{d_1} : \|z\| \leq 1\}$ and a function $G : (0,\infty) \times \mathcal{B}_{d_1}(1) \to \mathbb{R}$ as,

$$G(y, w) := \int_{\|z\|\leq y} \frac{1 - \cos\langle w, z\rangle}{\|z\|^{1+d_1}}\, dz. \tag{G.5}$$

By [50, Lemma 1], $G(\cdot,\cdot)$ is uniformly bounded, and by an application of the dominated convergence theorem, $\lim_{\delta\downarrow 0} G(y, w) = 0$, for each $w \in \mathcal{B}_{d_1}(1)$. Next note that, for any $c > 1$,

$$|W_{n,c} - D_n| \leq \int_{\{\|t\|\leq 1/c\}\cup\{\|t\|\geq c\}} |\xi_n(t,s)|^2 w(t,s)\, dt\, ds$$
$$+ \int_{\{\|s\|\leq 1/c\}\cup\{\|s\|\geq c\}} |\xi_n(t,s)|^2 w(t,s)\, dt\, ds. \tag{G.6}$$

We will use $\lesssim$ to hide constants which depend only on d_1 and d_2. Therefore,

$$\mathbb{E}\int_{\|t\|\leq 1/c} |\xi_n(t,s)|^2 w(t,s)\, dt\, ds$$
$$\overset{(a)}{\lesssim} \frac{n}{n-1} \int_{\|t\|\leq 1/c} \frac{(1 - |f_U^n(t)|^2)(1 - |f_V^n(s)|^2)}{\|t\|^{1+d_1}\|s\|^{1+d_2}}\, dt\, ds$$
$$\overset{(b)}{=} \frac{n}{n-1} \cdot \frac{1}{n^4} \sum_{k,l,m,h} \int_{\|t\|\leq 1/c} \frac{1 - \cos\langle t, U_k - U_l\rangle}{\|t\|^{1+d_1}} \cdot \frac{1 - \cos\langle s, V_m - V_h\rangle}{\|s\|^{1+d_2}}\, dt\, ds$$
$$\overset{(c)}{\lesssim} \frac{n}{n-1} \cdot \frac{1}{n^4} \sum_{k,l,m,h} G\left(\frac{\|U_k - U_l\|}{c}, \frac{U_k - U_l}{\|U_k - U_l\|}\right) \cdot \|V_m - V_h\| \tag{G.7}$$

where (a) follows from Fubini's Theorem and the calculations from (G.1), (b) uses the fact that $\sin(\cdot)$ is an odd function and hence integrates to 0 when integrated over symmetric sets, (c)

uses the definition from (G.5) and [50, Lemma 1]. The right hand side of (G.7) converges to

$$\mathbb{E}\left[G\left(\frac{\|\widetilde{U}_1-\widetilde{U}_2\|}{c},\frac{\widetilde{U}_1-\widetilde{U}_2}{\|\widetilde{U}_1-\widetilde{U}_2\|}\right)\right]\cdot\mathbb{E}\left[\|\widetilde{V}_1-\widetilde{V}_2\|\right] \tag{G.8}$$

where $\widetilde{U}_1,\widetilde{U}_2\sim\mathcal{U}^{d_1}$ and $\widetilde{V}_1,\widetilde{V}_2\sim\mathcal{U}^{d_2}$ are four independent random variables. Finally, by an application of the dominated convergence theorem, (G.8) converges to 0 as $c\to\infty$. By the same calculation as in (G.8), we get:

$$\begin{aligned}\mathbb{E}\int_{\|t\|\geq c}|\xi_n(t,s)|^2w(t,s)\,dt\,ds &\lesssim \frac{n}{n-1}\cdot\frac{1}{n^4}\sum_{k,l,m,h}\|V_m-V_h\|\cdot\int_{\|t\|\geq c}\frac{dt}{\|t\|^{1+d_1}}\\ &\lesssim\int_{\|t\|\geq c}\frac{dt}{\|t\|^{1+d_1}}.\end{aligned} \tag{G.9}$$

Clearly, the right hand side of (G.9) converges to 0 as limits are taken over $n\to\infty$ followed by $c\to\infty$. We can use the same arguments from (G.7) and (G.9) on the second term in the right hand side of (G.6) to get the same conclusion. Therefore, by an application of Markov's inequality, for any $\epsilon>0$,

$$\lim_{c\to\infty}\limsup_{n\to\infty}\mathbb{P}[|W_{n,c}-D_n|>\epsilon]\leq\lim_{c\to\infty}\limsup_{n\to\infty}\frac{1}{\epsilon}\cdot\mathbb{E}[|W_{n,c}-D_n|]=0.$$

This completes the proof. □

Proof of Proposition G.1. This proof is exactly the same as that of Lemma G.1 and we leave the details to the reader. One can also use the tightness of D (see (F.26)) as shown in [32, Chapter 1, Section 2]. □

G.2 Proof of Lemma F.2

Note that, for any $a\in\mathcal{S}^{d-1}$ and $U_i\in[0,1]^d$, $|a^\top U_i|\leq\|U_i\|\leq\sqrt{d}$. Therefore,

$$E_{m,n}=\int_{\mathcal{S}^{d-1}}\int_{-\sqrt{d}}^{\sqrt{d}}\Theta^2_{m,n}(a,r)\,dr\,d\kappa(a)\ \text{ and }\ E=\int_{\mathcal{S}^{d-1}}\int_{-\sqrt{d}}^{\sqrt{d}}\Theta^2(a,r)\,dr\,d\kappa(a).$$

From Lemma G.3 and Corollary G.1, we have convergence (weakly and in second moments) of the finite dimensional distributions of the process $\Theta_{m,n}(a,r)$, $a\in\mathcal{S}^{d-1}$, $r\in[-\sqrt{d},\sqrt{d}]$. Next

note that $\Theta_{m,n}(a,r)$ may be rewritten as,

$$\Theta_{m,n}(a,r) = \sqrt{\frac{n}{m}} \cdot \sqrt{m+n} \left(\frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(a^\top U_i \le t) - \frac{1}{n} \sum_{j=m+1}^{m+n} \mathbf{1}(a^\top U_{\pi_N(j)} \le t) \right).$$

Further observe that the set $\mathcal{F} := \{\mathbf{1}(a^\top \cdot \le t) : (a,t) \in \mathcal{S}^{d-1} \times [-\sqrt{d}, \sqrt{d}]\}$ of indicator functions on closed half-spaces is a VC class with index $d+1$ (see e.g., [14]) and consequently satisfies the uniform entropy condition, as in [52, Equation 2.5.1] (see e.g., [52, Theorem 2.6.7]). The asymptotic equicontinuity of the process $\Theta_{m,n}(a,r)$ over $\mathcal{S}^{d-1} \times [-\sqrt{d}, \sqrt{d}]$ then follows from the same proof as in [52, Theorem 2.5.2] (as it uses similar empirical process tools as in the proof of Lemma F.1, we leave the details to the interested reader). This then implies that $\Theta_{m,n}(\cdot,\cdot)$ converges weakly to $\Theta(\cdot,\cdot)$ in $L^\infty\big(\mathcal{S}^{d-1} \times [-\sqrt{d}, \sqrt{d}]\big)$ (see [27]). The weak convergence of $E_{m,n}$ to E then follows from a direct application of the continuous mapping theorem.

Lemma G.3. *Recall the notation and assumptions introduced in Lemma F.2. Consider a K-tuple, $(a_1,r_1),\ldots,(a_K,r_K)$, where $(a_i,r_i) \in \mathcal{S}^{d-1} \times \mathbb{R}$. Then the vector $\big(\Theta_{m,n}(a_1,r_1),\ldots,\Theta_{m,n}(a_K,r_K)\big)$ converges weakly to a multivariate Gaussian distribution with mean 0 and covariance matrix $\Sigma_{K\times K}$, where $\Sigma_{ij} = C\big((a_i,r_i),(a_j,r_j)\big)$ (given in (F.42)), as $\min(m,n) \to \infty$.*

Proof. For the sake of simplicity, we will work with $K=2$. Set $\alpha^\top := (\alpha_1,\alpha_2) \in \mathbb{R}^2$ and $\Theta_{m,n}^\top := \big(\Theta_{m,n}(a_1,r_1), \Theta_{m,n}(a_2,r_2)\big)$. It suffices to show (by the Cramér-Wold Theorem), $\alpha^\top \Theta_{m,n} \xrightarrow{w} \mathcal{N}(0, \alpha^\top \Sigma \alpha)$ as $\min(m,n) \to \infty$. Our proof proceeds using Stein's method of exchangeable pairs, see e.g., [10]. For the reader's convenience, we also present this result in Proposition H.2. Define $T_{m,n} := \alpha^\top \Theta_{m,n}$. Draw two random indices I and J, without replacement, from the set $\{1,2,\ldots,N\}$. Construct a new permutation, $\widetilde{\pi}_N$ as $\widetilde{\pi}_N(I) = \pi_N(J)$, $\widetilde{\pi}_N(J) = \pi_N(I)$ and $\widetilde{\pi}_N(k) = \pi_N(k)$ for $k \ne I, J$. It is easy to check that $(\pi_N, \widetilde{\pi}_N)$ forms an exchangeable pair of random vectors. Let $\widetilde{T}_{m,n} := \alpha^\top \widetilde{\Theta}_{m,n}$ where $\widetilde{\Theta}_{m,n}$ is calculated by replacing π_N with $\widetilde{\pi}_N$ in $\theta_{m,n}$. Note that,

$$\begin{aligned}
&\mathbb{E}[T_{m,n} - \widetilde{T}_{m,n} | \pi_N] \\
&= \mathbb{E}\Bigg[\frac{2\alpha_1\sqrt{m+n}}{\sqrt{mn}} \Big(\mathbf{1}(a_1^\top U_{\pi_N(I)} \le r_1) - \mathbf{1}(a_1^\top U_{\pi_N(I)} \le r_1) \Big) \mathbf{1}(I \le m, J \ge m+1) \\
&+ \frac{2\alpha_2\sqrt{m+n}}{\sqrt{mn}} \Big(\mathbf{1}(a_2^\top U_{\pi_N(I)} \le r_2) - \mathbf{1}(a_2^\top U_{\pi_N(I)} \le r_2) \Big) \mathbf{1}(I \le m, J \ge m+1) \Bigg| \pi_N \Bigg]
\end{aligned}$$

$$= 2(m+n-1)^{-1}T_{m,n} \tag{G.10}$$

which in turn implies, $\mathbb{E}[T_{m,n} - \widetilde{T}_{m,n}|T_{m,n}] = 2(m+n-1)^{-1}T_{m,n}$. Define $c_0 \coloneqq (m+n-1)(2\alpha^\top\Sigma\alpha)^{-1}$. Note that $|T_{m,n} - \widetilde{T}_{m,n}| \le 2(|\alpha_1| + |\alpha_2|)(\min(m,n))^{-1/2}$. By [10, Theorem 1.2], our desired conclusion follows if we can show the following:

$$\mathbb{E}\left|1 - \frac{c_0}{2}\mathbb{E}\left[(T_{m,n} - \widetilde{T}_{m,n})^2 \middle| T_{m,n}\right]\right| \overset{\min(m,n)\to\infty}{\longrightarrow} 0. \tag{G.11}$$

Note that,

$$\begin{aligned}
&\mathbb{E}\left[\frac{c_0}{2}\cdot\left(T_{m,n} - \widetilde{T}_{m,n}\right)^2 \middle| \pi_N\right] \\
&= \frac{(m+n)(m+n-1)}{(2\alpha^\top\Sigma\alpha)(mn)}\mathbb{E}\Bigg[\Bigg\{\alpha_1\Big(\mathbf{1}(a_1^\top U_{\pi_N(I)} \le r_1) - \mathbf{1}(a_1^\top U_{\pi_N(J)} \le r_1)\Big)\mathbf{1}(I \le m, \\
&J \ge m+1) + \alpha_2\Big(\mathbf{1}(a_2^\top U_{\pi_N(I)} \le r_2) - \mathbf{1}(a_2^\top U_{\pi_N(J)} \le r_2)\Big)\mathbf{1}(I \le m, J \ge m+1)\Bigg\}^2 \Bigg| \pi_N\Bigg] \\
&= \frac{\alpha_1^2}{(mn)(2\alpha^\top\Sigma\alpha)}\Bigg[n\sum_{i=1}^{m}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1) + m\sum_{j=m+1}^{m+n}\mathbf{1}(a_1^\top U_{\pi_N(j)} \le r_1) \\
&- 2\sum_{i\le m, j\ge m+1}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1, a_1^\top U_{\pi_N(j)} \le r_1)\Bigg] + \frac{\alpha_2^2}{(mn)(2\alpha^\top\Sigma\alpha)}\Bigg[n\sum_{i=1}^{m}\mathbf{1}(a_2^\top U_{\pi_N(i)} \le r_2) \\
&+ m\sum_{j=m+1}^{m+n}\mathbf{1}(a_2^\top U_{\pi_N(j)} \le r_2) - 2\sum_{i\le m, j\ge m+1}\mathbf{1}(a_2^\top U_{\pi_N(i)} \le r_2, a_2^\top U_{\pi_N(j)} \le r_2)\Bigg] \\
&+ \frac{\alpha_1\alpha_2}{(mn)(2\alpha^\top\Sigma\alpha)}\Bigg[n\sum_{i=1}^{m}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1, a_2^\top U_{\pi_N(i)} \le r_2) + m\sum_{j=m+1}^{m+n}\mathbf{1}(a_1^\top U_{\pi_N(j)} \le r_{,}1 \\
&a_2^\top U_{\pi_N(j)} \le r_2) - 2\sum_{i\le m, j\ge m+1}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1, a_2^\top U_{\pi_N(j)} \le r_2)\Bigg].
\end{aligned} \tag{G.12}$$

Further $\mathbb{E}[m^{-1}\sum_{i\le m}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1)] = N^{-1}\sum_{i\le N}\mathbf{1}(a_1^\top U_i \le r_1) \to \mathbb{P}(a_1^\top\widetilde{U} \le r_1)$ and $\mathrm{Var}[m^{-1}\sum_{i\le m}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1)] = \mathcal{O}(m^{-1})$. Therefore, $m^{-1}\sum_{i\le m}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1) \overset{\mathbb{P}}{\longrightarrow} \mathbb{P}(a_1^\top\widetilde{U} \le r_1)$. Similar arguments may be used to prove that

$$\begin{aligned}
&\frac{1}{m}\sum_{i=1}^{m}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1, a_2^\top U_{\pi_N(i)} \le r_2) \to \mathbb{P}(a_1^\top\widetilde{U} \le r_1, a_2^\top\widetilde{U} \le r_2) \qquad \text{and} \\
&\frac{1}{mn}\sum_{i\le m, j\ge m+1}\mathbf{1}(a_1^\top U_{\pi_N(i)} \le r_1, a_2^\top U_{\pi_N(j)} \le r_2) \to \mathbb{P}(a_1^\top\widetilde{U} \le r_1)\mathbf{P}(a_2^\top\widetilde{U} \le r_2).
\end{aligned} \tag{G.13}$$

Recall the definition of $C(\cdot,\cdot)$ from (F.42). Note that (G.13) implies (G.12) converges in

probability to

$$\frac{1}{\alpha^\top \Sigma \alpha}\left[\alpha_1^2 C((a_1,r_1),(a_1,r_1)) + 2\alpha_1\alpha_2 C((a_1,r_1),(a_2,r_2)) + \alpha_2^2 C((a_2,r_2),(a_2,r_2))\right] = 1. \quad \text{(G.14)}$$

Finally, as $c_0(T_{m,n} - \widetilde{T}_{m,n})^2$ is uniformly bounded, (G.14) implies (G.11) by the dominated convergence theorem. □

Corollary G.1. *Recall the notation from the statement and proof of Lemma G.3. Then* $\mathbb{E}[T_{m,n}^2] \to \alpha^\top \Sigma \alpha$ *as* $\min(m,n) \to \infty$.

Proof. In the proof of Lemma G.3, we showed that $(c_0/2)\mathbb{E}(T_{m,n} - \widetilde{T}_{m,n})^2 \to 1$. Considering all limits to be under $\min(m,n) \to \infty$, we get:

$$\begin{aligned} 1 = \lim (c_0/2)\mathbb{E}(T_{m,n} - \widetilde{T}_{m,n})^2 &= \lim c_0 \mathbb{E}\left[T_{m,n}\mathbb{E}[T_{m,n} - \widetilde{T}_{m,n} | T_{m,n}]\right] \\ &\overset{(a)}{=} \lim 2c_0(m+n-1)^{-1}\mathbb{E}[T_{m,n}^2] \end{aligned}$$

which completes the proof. Here, (a) follows from (G.10). □

H Auxiliary Results

Lemma H.1 (Alexandroff Theorem, Alexandroff [2])**.** *Let* $f : U \to \mathbb{R}$ *be a convex function, where* U *is an open convex subset of* $\mathbb{R}^n$*. Then* f *has a second derivative Lebesgue a.e. in* U*.*

Lemma H.2. *(Almost sure weak convergence of empirical measure, Varadarajan [53]) Let* (W,d) *be a separable metric space and* μ *be a probability measure supported on* W*. Also, say* μ_n *denotes the empirical counterpart of* μ*. Then* $d_W(\mu_n,\mu) \xrightarrow{a.s.} 0$ *where* $d_W(\cdot,\cdot)$ *is any metric on the space of probability measures on* (W,d) *that equivalently characterizes weak convergence.*

Lemma H.3. *(McCann [34, Lemma 9]) Suppose* $\mu_n \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ *converges weakly to* $\mu \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$*. Then,*

(i) If μ_n *has cyclically monotone support for all large* n*, then so does* μ*.*

(ii) Let $\Gamma(\nu_1,\nu_2)$ *denote the subset of* $\mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ *with first and second marginals* ν_1 *and* ν_2 *respectively;* $\nu_1,\nu_2 \in \mathcal{P}(\mathbb{R}^d)$*. If* $\mu_n \in \Gamma(\nu_1^n,\nu_2^n)$ *where* $\nu_1^n \xrightarrow{w} \nu_1$ *and* $\nu_2^n \xrightarrow{w} \nu_2$*, then* $\mu \in \Gamma(\nu_1,\nu_2)$*.*

Definition H.1 (Cyclically monotone maps)**.** *A subset S of $\mathbb{R}^d \times \mathbb{R}^d$ is said to be cyclically monotone if, given any finite subset of S, say $\{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_k, \mathbf{y}_k)\}$, we have:*

$$\langle \mathbf{y}_1, \mathbf{x}_2 - \mathbf{x}_1 \rangle + \ldots + \langle \mathbf{y}_{k-1}, \mathbf{x}_k - \mathbf{x}_{k-1} \rangle + \langle \mathbf{y}_k, \mathbf{x}_1 - \mathbf{x}_k \rangle \leq 0.$$

A multi-valued map $f : \mathbb{R}^d \to \mathbb{R}^d$ is said to be a cyclically monotone map if, given any finite subset $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ of $\mathbb{R}^d$, the set $\{(\mathbf{x}_1, f(\mathbf{x}_1)), \ldots, (\mathbf{x}_k, f(\mathbf{x}_k))\}$ is cyclically monotone.

Definition H.2 (Subdifferential of a convex function)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be a proper, lower semicontinuous convex function. Then the subdifferential of $f(\cdot)$ at a point $\mathbf{x} \in \mathbb{R}^d$ is defined as:*

$$\partial f(\mathbf{x}) \coloneqq \{\mathbf{z} : f(\mathbf{y}) - f(\mathbf{x}) \geq \langle \mathbf{z}, \mathbf{y} - \mathbf{x} \rangle, \quad \textit{for all } \mathbf{y} \in \mathbb{R}^d\}.$$

Lemma H.4. *(Cyclic monotonicity and subdifferential of convex functions; Rockafellar [44, Theorem 1]) The graph of the subdifferential $\partial f(\cdot)$ of a convex function $f : \mathbb{R}^d \to \mathbb{R}$ is a cyclically monotone subset of $\mathbb{R}^d \times \mathbb{R}^d$. Moreover, any cyclically monotone subset of $\mathbb{R}^d \times \mathbb{R}^d$ is contained in the graph of the subdifferential of a proper, lower semicontinuous convex function from $\mathbb{R}^d \to \mathbb{R}$.*

Lemma H.5. *(Uniqueness of measure preserving couplings, see McCann [34, Corollary 14]) Let $\nu_1, \nu_2 \in \mathcal{P}(\mathbb{R}^d)$, and suppose that one of these two measures in Lebesgue absolutely continuous. Then, there exists one and only one measure $\nu \in \Gamma(\nu_1, \nu_2)$ (see (ii) from Lemma H.3) with cyclically monotone support.*

Lemma H.6. *(Existence of measure transformation maps; see [34, Proposition 10]) Assume that $\nu \in \Gamma(\nu_1, \nu_2)$ (see (ii) from Lemma H.3) is supported on the graph of the subdifferential $\partial f(\cdot)$ of some proper, lower semicontinuous convex function $f : \mathbb{R}^d \to \mathbb{R}$ (i.e., the support of ν is a subset of the graph of $\partial f(\cdot)$). Further, suppose that $\nu_1 \in \mathcal{P}_{ac}(\mathbb{R}^d)$. Then $\nabla f(\cdot)$ pushes ν_1 to ν_2, i.e., $\nu = (\textit{identity} \times \nabla f)\#\nu_1$.*

Proposition H.1. *(Hoeffding's Central Limit theorem; see Chen and Fang [11, Theorem 1.1]) Suppose $\mathfrak{X} = \{X_{ij} : 1 \leq i, j \leq n\}$ be a $n \times n$ array of independent random variables where*

$n \geq 2$, $\mathbb{E}[X_{ij}] = c_{ij}$, $\mathrm{Var}(X_{ij}) = \sigma_{ij}^2 \geq 0$ *and* $\mathbb{E}|X_{ij}|^3 < \infty$. *Assume that,*

$$c_{i\cdot} := \frac{1}{n}\sum_j c_{ij} = 0 \qquad \textit{and} \qquad c_{\cdot j} := \frac{1}{n}\sum_i c_{ij} = 0.$$

Let π_n *be an uniform permutation drawn from* S_n *(all permutations of* $\{1, 2, \ldots, n\}$*) independent of* $\mathfrak{X}$. *Let* $W_n = \sum_i X_{i\pi_n(i)}$. *Then,*

$$\mathrm{Var}(W_n) = \frac{1}{n}\sum_{i,j}\sigma_{ij}^2 + \frac{1}{n-1}\sum_{i,j} c_{ij}^2.$$

Further, if $\mathrm{Var}(W_n) = 1$, *then we have:*

$$\sup_{z\in\mathbb{R}}\Big|\mathbb{P}(W_n < z) - \Phi(z)\Big| \leq \frac{451}{n}\sum_{i,j}\mathbb{E}|X_{ij}|^3$$

where $\Phi(\cdot)$ *denotes the standard Gaussian distribution function.*

Proposition H.2. *(Stein's method of exchangeable pairs; see Chatterjee and Shao [10, Theorem 1.2]) Let* (W, W') *denote an exchangeable pair of random variables where* W *has finite variance. Also, suppose that,*

$$\mathbb{E}(W - W'|W) = g(W) + r(W) \qquad \text{and} \qquad |W - W'| \leq \delta,$$

where δ *is a constant (non-random) and* $g(\cdot)$ *is a differentiable function on* $\mathbb{R}$. *Let* $G(t) = \int_0^t g(s)\,ds$ *and* $p(t) = c_1\exp(-c_0 G(t))$, *where* $c_1 = (\int \exp(-c_0 G(t))\,dt)^{-1} > 0$. *Here* c_0 *is some positive real number. Further, let us also assume the following conditions:*

(i) $g(\cdot)$ *is nondecreasing,* $g(t) \geq 0$ *for* $t \geq 0$ *and* $g(t) \leq 0$ *for* $t \leq 0$.

(ii) There exists $c_2 < \infty$ *such that for all* $x \in \mathbb{R}$,

$$\min\big(1/c_1, 1/|c_0 g(x)|\big)\big(|x| + 3/c_1\big)c_0|g'(x)| \leq c_2.$$

Finally, let Y be a random variable with density $p(\cdot)$. Under all the above conditions, the following holds:

$$\sup_{z\in\mathbb{R}} \big|\mathbb{P}(W \le z) - \mathbb{P}(Y \le z)\big| \le 3\big|1 - (c_0/2)\mathbb{E}[(W - W')^2|W]\big| + c_1 \max(1, c_2)\delta$$
$$+ 2(c_0/c_1)\mathbb{E}\big|r(W)\big| + \delta^3 c_0\big\{(2 + c_2)/2\mathbb{E}\big|c_0 g(W)\big| + c_1 c_2/2\big\}.$$

I R Pseudo-codes

With a slight notational abuse, $\mathbf{X}$ and $\mathbf{Y}$ here will denote the $n \times d_1$ and $n \times d_2$ data matrices comprising n vectors of dimensions d_1 and d_2 respectively. In other words, $\mathbf{X}_{n\times d_1} = [\mathbf{X}_1, \ldots, \mathbf{X}_n]^\top$ and $\mathbf{Y}_{n\times d_2} = [\mathbf{Y}_1, \ldots, \mathbf{Y}_n]^\top$. Ignoring the boldface notation, the pseudo-code is as follows:

```
##### Pearson's test #####
cor.test(X,Y,method="pearson")$p.value

##### Distance correlation test #####
dcor.test(X,Y,R=500)$p.val

##### HSIC test #####
dhsic.test(X,Y)$p.val

##### MINT #####
MINTav(X,Y,K=1:20)[1]
MINTauto(X,Y,kmax=50)[1]
```

```
##### HHG #####
hhg.test(X,Y,nr.perm = 1000)$perm.pval.hhg.sl
```

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